MES4-MCU · design review package R3

Input for schematic review R3 · schematic Rev A-d6 (title block 14 Oct 2026; notes updated 26 and 29 Oct) · 29 Oct 2026 · Monkey Robotics · Aaron Raj Bhatti

What's in this package: (1) this cover: what changed since review R2, the status of every R2 finding and what to look at; (2) the LTspice simulation report, 10 benches plus a model check, grouped by schematic sheet; (3) the 5-page schematic (A3), appended after the report. The carrier interface specification (MES4-CIF) is referenced but not included. Its must-follow rules are printed on schematic sheet 4.

The module: STM32G0B1KBT6 on an M.2 2242 B-key card (custom pinout, not a PC M.2 card), 5 V in, TLV76733 3.3 V LDO, SN65HVD230 CAN at 500 kbit/s, 4 status LEDs, passive buzzer and a 5 V monitor. Rev A is not fabricated yet.

A · Changes since review R2 (Rev A-d4, 8 Oct)

Sheet (page)	Change	Why
Power (2)	**BZ1 → KLJ-5018-5040** passive magnetic buzzer (5.5 × 5.5 × 1.8 mm, 25 Ω), driven with 4 kHz PWM on PB7 (TIM4_CH2) through Q3. Idle LOW, never DC	d5, 13 Oct: smaller part
MCU (3)	**U1 STM32G0B1KBU6 (UFQFPN-32) → STM32G0B1KBT6 (LQFP-32)**, non-N pinout. No exposed pad: pin 5 is the only VSS	d6, 14 Oct: KBU6 out of stock (the sheet 1 revision block also notes "review fixes" for d6)
MCU (3)	Exposed-pad notes corrected for the LQFP-32: pin 5 runs to the C4/C5 ground pads and the GND plane	26 Oct: the notes still described the UFQFPN pad
MCU (3)	NRST note: C6 recharges in **≈ 5 ms (3–7 ms)**, was ≈ 4 ms	29 Oct: sim MCU2 (25–55 kΩ pull-up range)
CAN_M2 (4)	Carrier rule 4: **+5V never below −0.3 V** at the fingers	29 Oct: sim PWR3. A negative transient lets U2's body diode drag 3V3 below the MCU's limit
Docs	README: JLC panel needed for bevelled fingers; CAN must not share a cable with > 12 V. BRINGUP step 12: buzzer ≤ 100 mA extra (was 80)	26 / 29 Oct; sim BZ1
sim/ (new)	LTspice benches for every analog and power circuit, fed by the real MES4-Carrier-EC 5 V path	Simulation evidence (report follows)

> The R2 decisions still stand: D1 TLV76733 LDO, D2 SN65HVD230 with its hardening (R7 0603, R15 pull-up on CAN_TX, R16 Rs link, rule 9), D3 no M.2 pinout change, D4 USB unused.

B · Status of the R2 findings

ID	Finding	Status	How / evidence
R1	HT7333-2 below its recommended input range	Fixed	TLV76733 (VIN 2.5–16 V). PWR4: 3V3 stays ≥ 3.25 V from a 3.38 V input at 148 mA
R2	LDO has no thermal shutdown	Fixed	TLV76733: thermal shutdown at 180 °C plus fold-back to 0.15–0.35 A on a pin-4 short
R3	3V3 budget used the no-load SN65HVD230 current	Fixed	Budget is 148 mA with U3 driving 60 Ω. PWR4: U2 0.29 W at 5.25 V. PWR5: ±7 mV under CAN traffic plus a 50 mA step
R4	Carrier rule asked for series R on CAN	Fixed	Rule 8: TVS only, no series R. CAN2: 22 Ω per line drops VOD to 1.25 V, 47 Ω to 0.88 V (below 0.9 V)
R5	D2 footprint SOD-123F vs the SMF part	Fixed	`Diode_SMD:D_SMF`
R6	PLL input duty cycle margin	Firmware	PLLM = 2 (4 MHz PLL input). MCU1: PC14 duty 50.0 % with R14, no overshoot past the clamp diodes
R7	PB2 injection reasoning was wrong	Fixed	R8 = 20 kΩ (÷3), note rewritten. PWR3: PB2 peaks at 2.76 V at D2's rated pulse. PWR4: ADC codes 1965 / 2068 / 2172 as in BRINGUP
R8	NUP2105L doesn't protect U3 from DC bus faults	Accepted	SN65HVD230 kept (D2). Carrier rule 9 and the README: CAN never shares a cable or connector with > 12 V

C · What to review

- **The schematic as a whole** (the five A3 pages at the end), especially what changed in d5 / d6: the passive buzzer drive on Power and the LQFP-32 U1 on MCU.
- **Buzzer current** (report §3, BZ1). The bench gives 69–95 mA extra and 0.3–0.5 W in the coil while beeping, against the datasheet's 80 mA. Is the PWM drive with the D3 freewheel acceptable as is, or should the default duty be lower?
- **Negative +5V transients** (report §3, PWR3). The module relies on the carrier (rule 4) to keep +5V above −0.3 V. Is that acceptable, or should the module get its own protection?
- **Model assumptions** (report §7). U2 and U3 are behavioural models, and X1's edge rate and BZ1's inductance are assumed, not published. Check that the conclusions don't hang on them.

MES4-MCU · LTspice simulations

Schematic Rev A-d6 (14 Oct 2026) · simulated 29 Oct 2026 · Monkey Robotics · Aaron Raj Bhatti

Ten LTspice benches cover every analog and power circuit on the module, grouped by schematic sheet like the MES4-Carrier-EC benches, plus one bench that checks the U2 regulator model against TI's datasheet. The +5V benches are fed by the real MES4-Carrier-EC 5 V path (U1 URB2405S-6WR3, F2, FB1, C6–C9). No circuit fails. Three results changed a bring-up limit or a note (section 2, done 29 Oct).

1 · Summary

Sheet (page)	Bench	Test	Result
Power (2)	PWR1	Power-up on the carrier +5V (2 ms and 100 µs ramps)	Pass
Power (2)	PWR2	Module pushed into a live slot (MES4-CIF rule 5 broken)	Informative
Power (2)	PWR3	19.4 A 10/1000 µs pulse on +5V (D2 rating), both polarities	Pass + – is the carrier's job
Power (2)	PWR4	+5V swept 3.0–7.5 V: LEDs, VSENSE ADC code, U2 dropout, D2	Pass
Power (2) + CAN (4)	PWR5	+3V3 under CAN traffic and a 50 mA pin-4 step	Pass
Power (2)	BZ1	Buzzer, 4 kHz 50 % PWM	Pass, bring-up limit
MCU (3)	MCU1	8 MHz HSE into PC14, R14 0 vs 22 Ω	Pass
MCU (3)	MCU2	NRST: power-up, internal reset, button	Pass
CAN (4)	CAN1	500 kbit/s on a 10 m bus with two 0.3 m stubs	Pass
CAN (4)	CAN2	Carrier series resistors on CANH / CANL (MES4-CIF rule 8)	Rule confirmed
Models	MDL1	U2 TLV76733 model against TI Fig. 7-12	Fits

2 · Findings and changes made

- **Buzzer current (BZ1).** With the coil modelled as 25 Ω plus inductance, beeping adds **69–95 mA** to the module's input current, depending on the coil inductance (0.1–1 mH, not published). The Power notes and BRINGUP step 12 say "≤ 80 mA". The real figure should be lower, because the buzzer's mechanical resonance raises its impedance at 4 kHz and the model leaves that out, but 80–95 mA is still possible. **Done:** BRINGUP step 12 pass limit is now "≤ 100 mA extra". Keep tones short: the coil dissipates 0.3–0.5 W while it beeps.
- **Negative transients on +5V (PWR3).** When D2 conducts forward (+5V at −2.6 V at its 19.4 A rating), U2's body diode empties the 3V3 capacitors into it. 3V3 goes to **−1.7 V** for about 3 ms, while the STM32's absolute minimum is −0.3 V. This is MES4-CIF rule 4: reverse and surge protection is the carrier's job. The

MES4-Carrier-EC's +5V comes from an isolated converter through F2 and FB1, so it can't do this. **Done:** MES4-CIF rule 4 and the rules box on sheet CAN_M2 now say to keep +5V above −0.3 V at the fingers.

- **NRST release time (MCU2).** The MCU-sheet note says the pull-up "recharges C6 in ≈ 4 ms". The bench gives **4.9 ms** to VIH with the typical 40 kΩ, and 3.0–6.7 ms over the 25–55 kΩ range. **Done:** the MCU-sheet note and MES4-CIF §4a now say "≈ 5 ms (3–7 ms)". There's no functional impact.
- **Hot-plug (PWR2).** Nothing on the module is overstressed (+5V peaks at 5.09 V, so D2 stays off), but the contacts carry a **96 A** spike for under a microsecond and the carrier socket dips to 2.3 V for about 5 µs. Rule 5 (no hot-plug) stands as written. That dip, plus the fingers mating in random order, is reason enough.

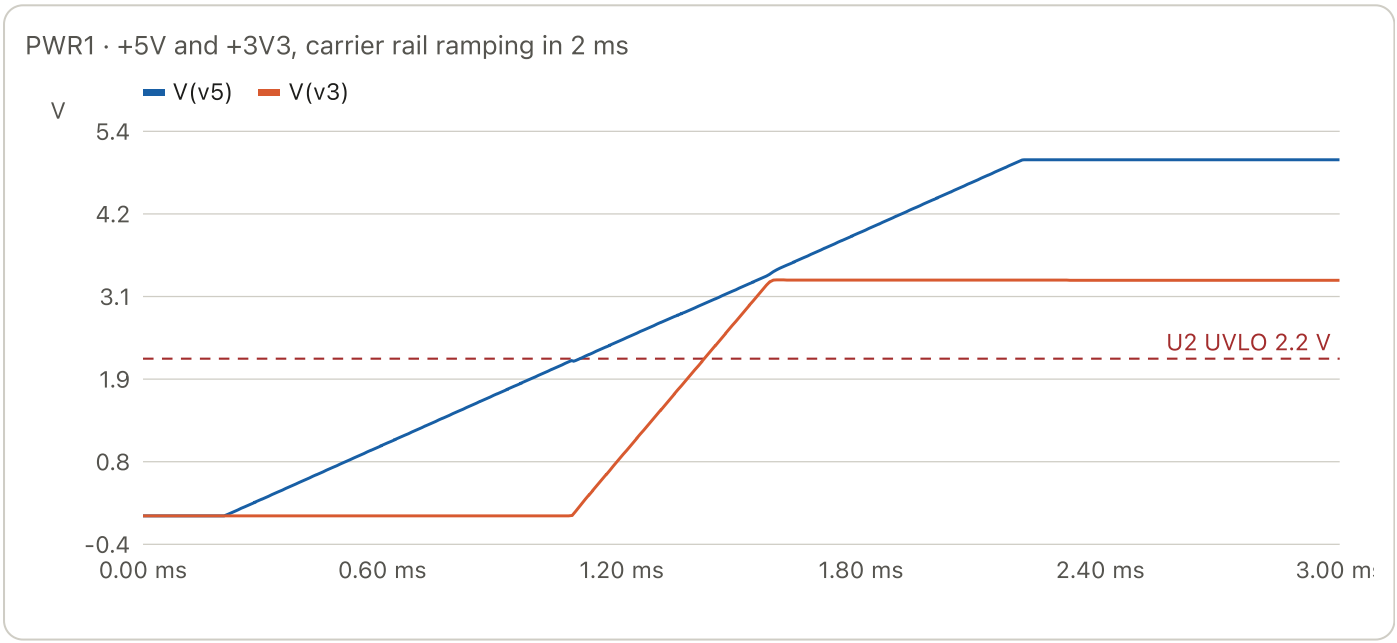
3 · Power · sheet Power (page 2)

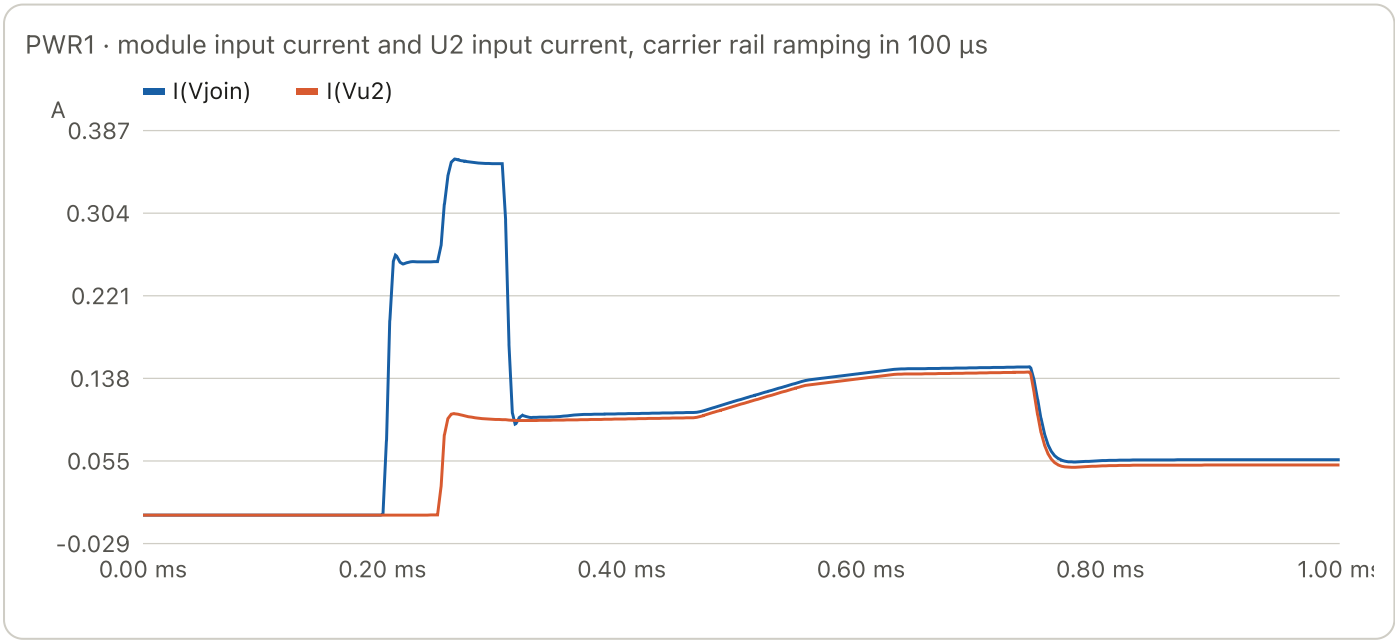
Supply: the MES4-Carrier-EC +5V path from its bench PWR5. U1 URB2405S-6WR3 output (0.1 Ω) → C6 22 µF / C7 → F2 1206L110 (0.12 Ω) → FB1 BLM21PG121SN1D → C8 10 µF / C9 at the socket. Model: `carrier5v.inc`.

Parts: fingers 20/22 (3 nH) · C1 10 µF, C10, D2 SMF6.0A · U2 TLV76733 · C2/C5 10 µF, C3/C4/C7/C8 100 nF · LED1–LED4 with R1–R4, Q1/Q2/Q4, R5/R6/R13 · R8/R9/C9 → PB2 · BZ1, Q3, D3, R12. Loads: STM32 20 mA, X1 10 mA, U3 10 mA (recessive). Model: `module.inc`.

PWR1 · power-up (`pwr1_powerup.net`)

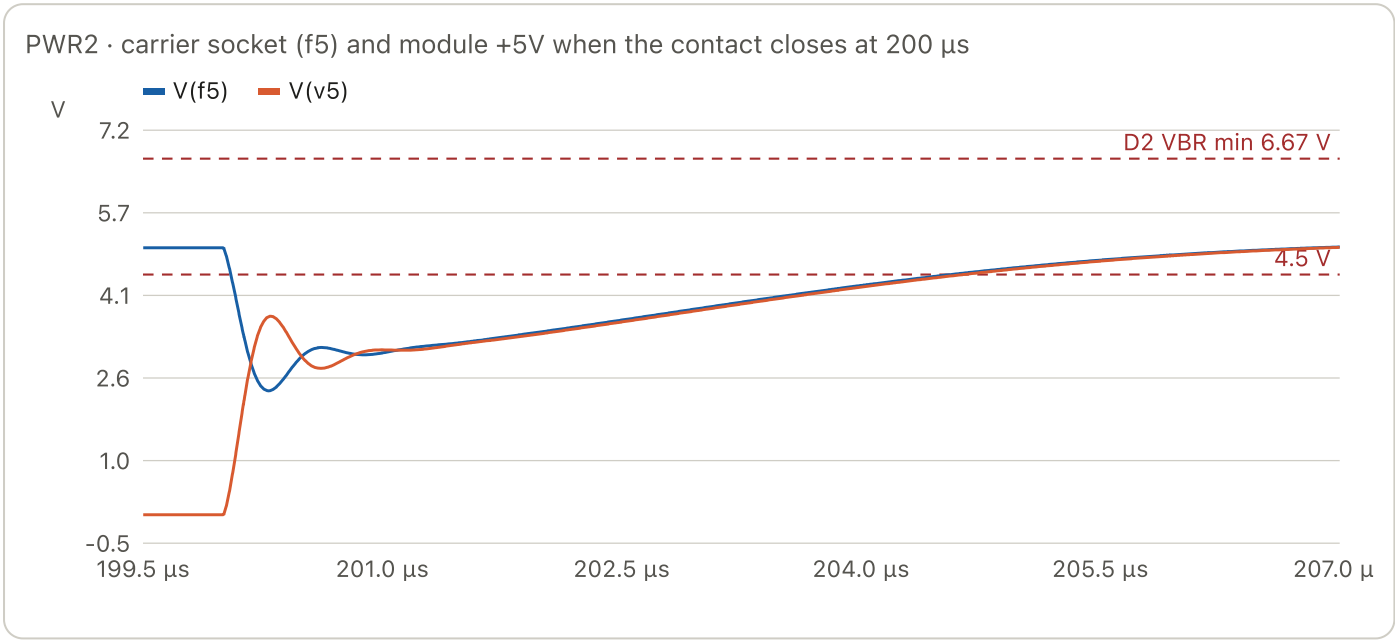
The carrier's U1 output ramps 0 → 5 V in 2 ms, or in 100 µs as a fast corner. U2 turns on when +5V passes its 2.2 V UVLO, and 3V3 reaches 3.0 V 0.45 ms later (500 µs soft-start). 3V3 doesn't overshoot (3.306 V max) and settles at 3.300 V. Peak input current is **0.16 A** with the 2 ms ramp and **0.36 A** with the 100 µs ramp, mostly C1 charging; U2's own inrush is held to 0.14 A by its soft-start. LED2 reaches 1 mA once +5V is above about 3.8 V and 3V3 above about 0.7 V (Q4's base), so it shows the rail is present, not that it's in regulation, as the notes say. Steady input current with these worst-case logic loads is 56 mA.

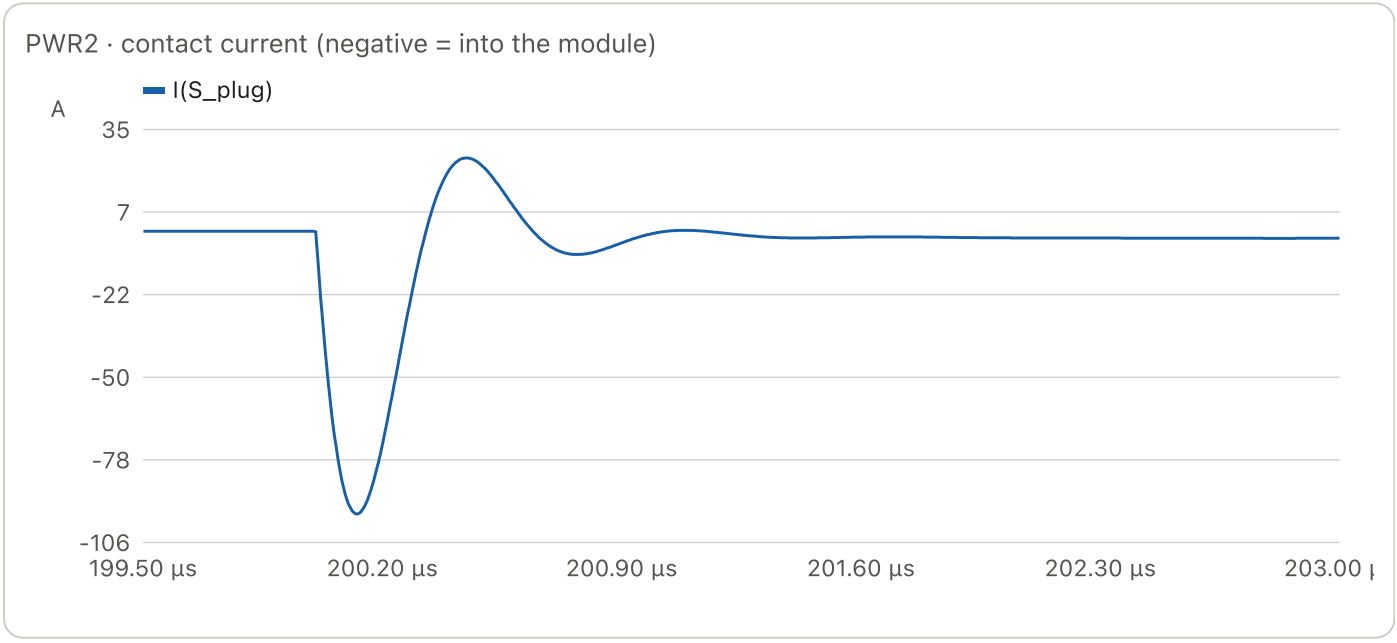




PWR2 · hot-plug, informative (`pwr2_hotplug.net`)

The module goes into a live slot: the carrier's C8 at the socket is at 5 V and the module's C1 is empty. The contact (10 mΩ, 3 nH, no bounce modelled) carries a **96 A** spike, rung out within about 1 µs: the two ceramics share 24 µC through almost no impedance, so the gold fingers will spark. The socket voltage drops to **2.3 V** and is back above 4.5 V within 4.5 µs. Module +5V peaks at 5.09 V, so D2 never reaches breakdown. The 20 mA it shows is its junction capacitance charging. U2 starts normally and 3V3 doesn't overshoot. Nothing here damages the module, but the spike and the dip land on everything else on the carrier's +5V. Back-powering through I/O fingers that mate before GND is not modelled. Rule 5 stays.

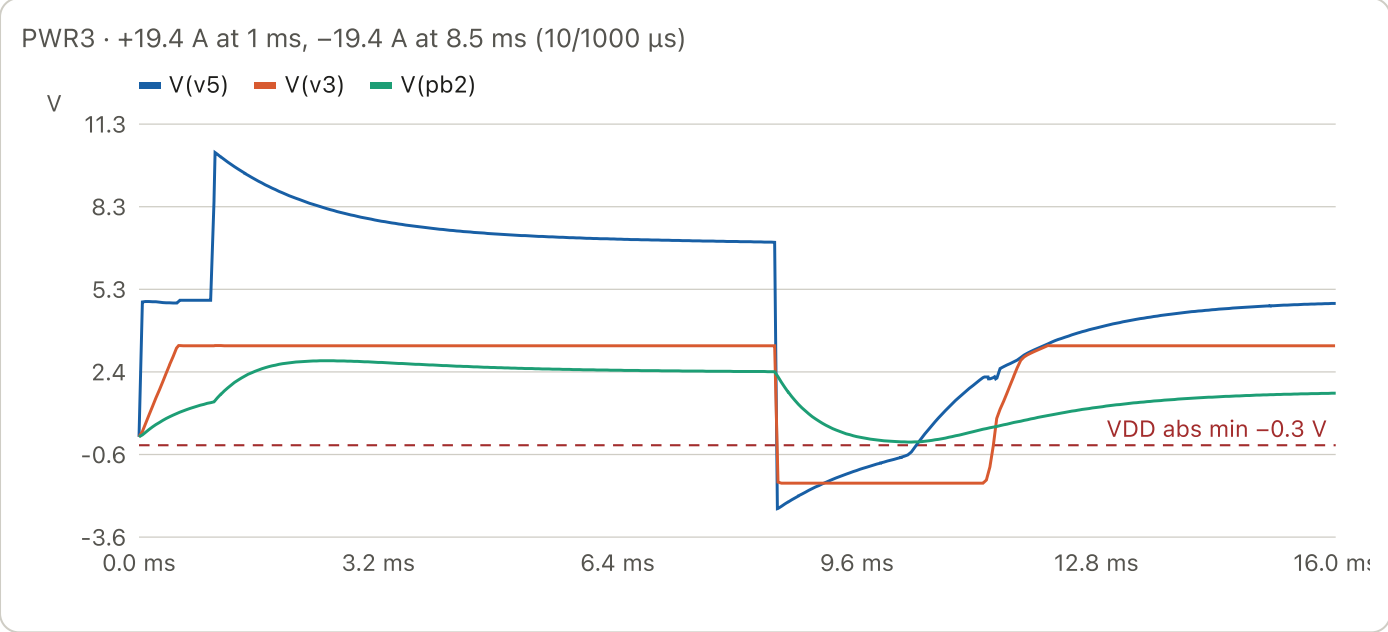




PWR3 · pulse at D2's rating (`pwr3_tvs_pulse.net`)

19.4 A, 10/1000 µs, injected straight at the fingers, positive at 1 ms and negative at 8.5 ms. The 5 V source here can't sink current, so D2 takes the whole positive pulse. **Positive:** +5V clamps at **10.3 V**, D2 carries 19.4 A and 0.24 J. U2 (18 V abs), C1 (25 V) and C10 (16 V) are fine. 3V3 doesn't move (3.306 V max). Because the source can't sink, the pulse's long tail keeps D2 clamping, and +5V sits near 7–8 V until 8.5 ms. PB2 peaks at 2.76 V: C9 filters out the brief 10.3 V peak, leaving 8.3 V / 3. That's below the 3.43 V worst case in the notes, and PB2 is FT_ea anyway. **Negative:** D2 conducts forward and +5V reaches −2.6 V. U2's body diode then discharges the

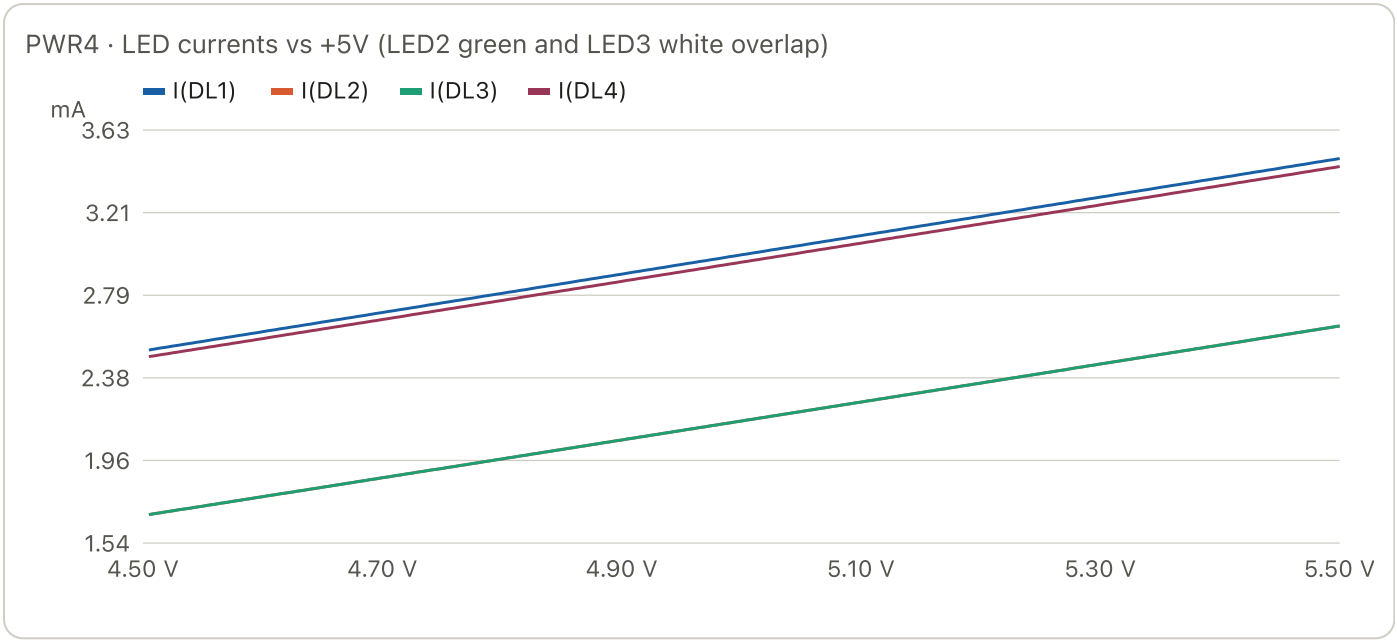
3V3 capacitors into the input, with up to 8.3 A flowing back through U2 in this model. 3V3 stays below −0.3 V from 8.5 to 11.4 ms and is back above 3.2 V at 12.1 ms. The MCU, X1 and U3 would be reverse-biased. See section 2: this is the carrier's job.

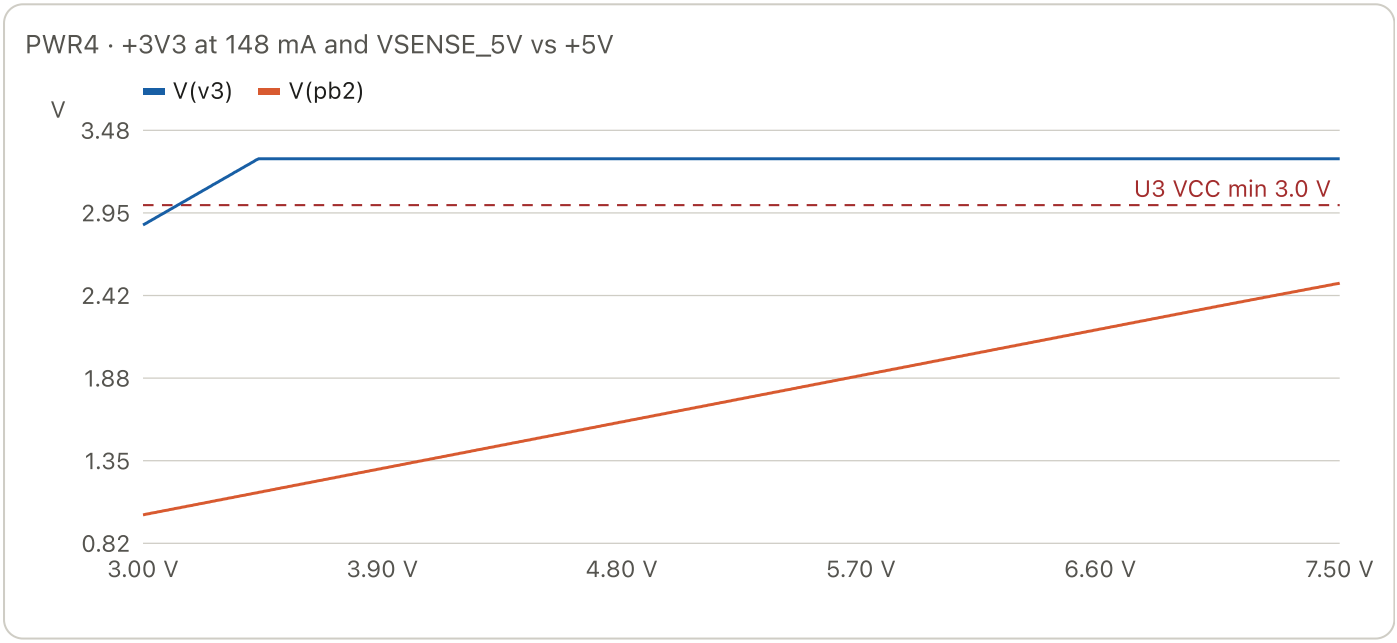


PWR4 · DC sweep of +5V (`pwr4_dc_sweep.net`)

+5V is swept from 3.0 to 7.5 V. The 3V3 load is the full 148 mA from the notes (MCU 20, U3 dominant into 60 Ω 67, X1 10, GPIO 1, pin 4 50 mA), with all four LEDs on.

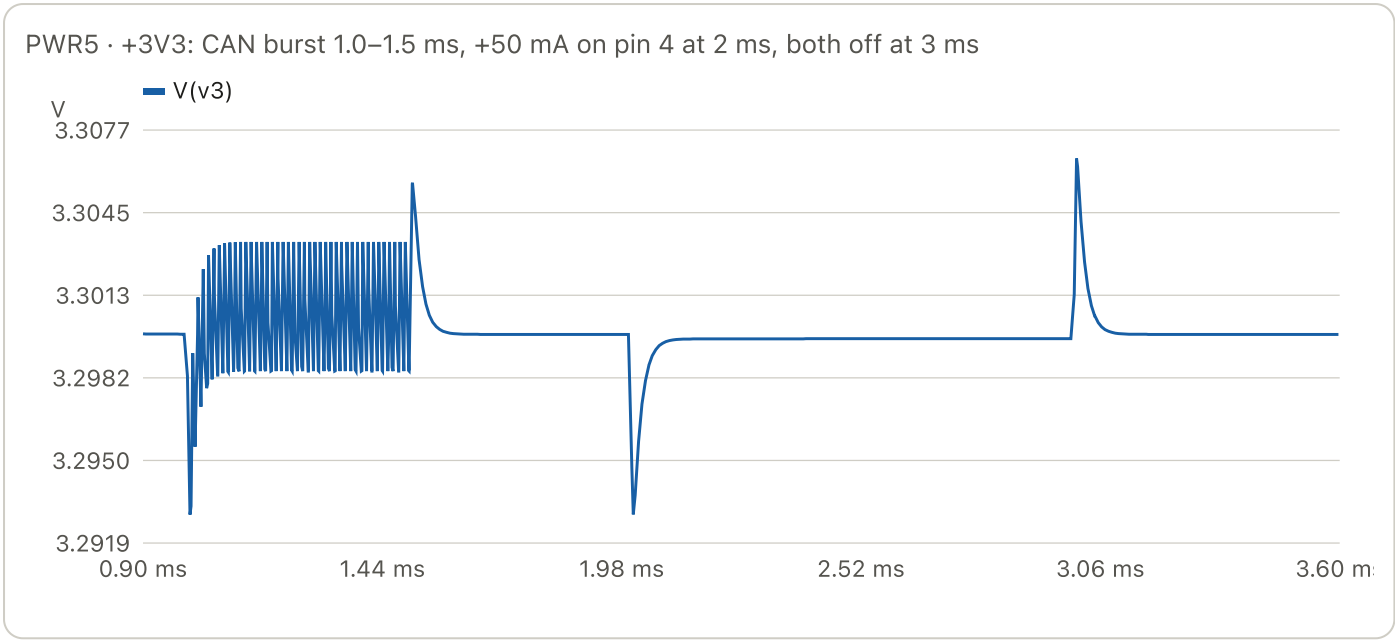
Check	Simulated	Notes / limit	Result
LED1 red (+5V)	2.76 mA at 4.75 V, 3.24 mA at 5.25 V	3.0 mA at 5.0 V	Pass
LED2 green (3V3, Q4)	2.16 mA	≈ 2.1 mA	Pass
LED3 white (CAN, Q1)	2.16 mA at 5.0 V, 1.92 mA at 4.75 V	≈ 2.1 mA	Pass
LED4 yellow (USR, Q2)	2.96 mA	≈ 2.9 mA	Pass
VSENSE ADC code (VDDA = real 3V3)	1965 / 2068 / 2172 at 4.75 / 5.00 / 5.25 V	BRINGUP step 10: 1965 / 2068 / 2172	Pass
3V3 in regulation (≥ 3.25 V) at 148 mA	from +5V = 3.38 V; 3.0 V (U3 VCC min) at 3.13 V	+5V min 4.75 V: 1.4 V of margin	Pass
Module input current, 148 mA logic + 4 LEDs	159 mA at 5.0 V	≤ 240 mA incl. buzzer (MES4-CIF rule 2)	Pass
U2 dissipation at 5.25 V, 148 mA	0.29 W	0.29 W → T_J 92 °C at 70 °C ambient	Pass
D2 onset (typical VBR 7.0 V model)	1 mA at 6.94 V, 10 mA at 7.00 V	VBR min 6.67 V at 10 mA; CIF max 5.5 V	Pass





PWR5 · 3V3 load steps (`pwr5_3v3_load.net`)

From 1.0 to 1.5 ms, CAN traffic at the bit-stuffing worst case (5 dominant / 1 recessive): U3 draws its 17 mA maximum plus 50 mA into the bus while dominant. At 2 ms the carrier adds 50 mA on pin 4 in 1 µs, and at 3 ms both loads switch off. 3V3 stays within **3.293–3.307 V** (±7 mV) throughout, and module +5V dips to 4.973 V at most. The U2 model is fitted to a 1 µF output, so the module's ≈ 14 µF (C2 + C5 + 4 × 100 nF) makes these figures conservative.



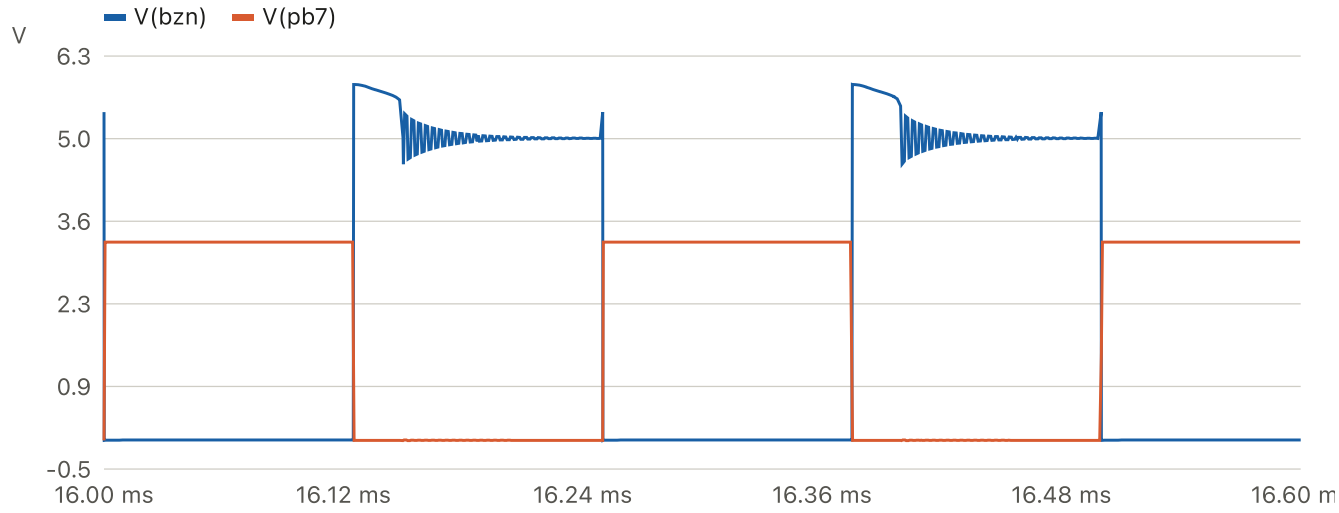
BZ1 · buzzer PWM (`bz1_buzzer_pwm.net`)

PB7 drives Q3 at 4 kHz, 50 %, for 10 ms. The coil is modelled as 25 Ω in series with 0.1, 0.3 or 1 mH (the inductance isn't published), with D3 freewheeling. Mechanical resonance is not modelled, so the currents are upper bounds.

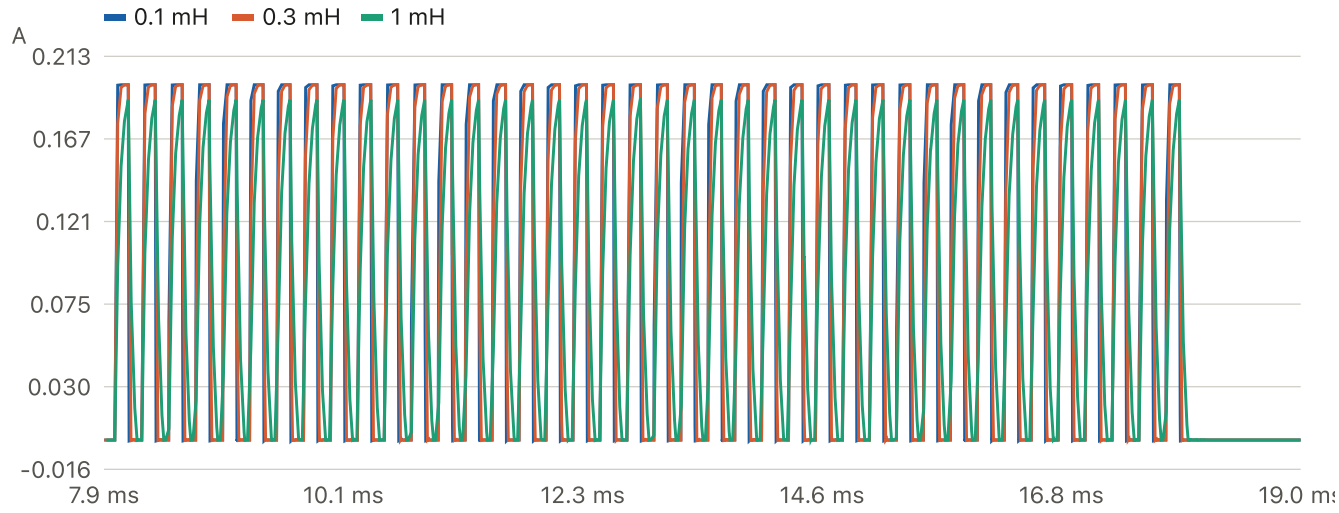
Coil inductance	0.1 mH	0.3 mH	1 mH	Limit
Extra module input current	95 mA	89 mA	69 mA	BRINGUP step 12: ≤ 100 mA (was 80)
Coil current: average / peak	98 / 197 mA	96 / 197 mA	89 / 189 mA	200 mA DC if PB7 sticks HIGH
Coil dissipation while beeping	0.47 W	0.43 W	0.32 W	1 W DC (note: never DC)
Q3 drain peak	5.87 V	5.87 V	5.89 V	AO3400A 30 V
+5V ripple at the module	53 mV p-p	53 mV p-p	51 mV p-p	4.75 V min
+3V3 ripple	72 µV p-p	66 µV p-p	54 µV p-p	—
VSENSE (PB2): offset / ripple	−10.1 / 1.9 LSB	−9.4 / 1.8 LSB	−7.2 / 1.4 LSB	firmware rule F5

The circuit is fine: Q3 sees under 6 V and D3's recovery spike is only 3 mA. The current is the item to watch (section 2). Firmware rule F5 (read VSENSE with the buzzer silent) is right, but the effect is small. While the buzzer runs, the reading drops by about 9 LSB (23 mV on the 5 V reading) plus 2 LSB of ripple. That drop is real: +5V sags across the carrier's F2 and FB1.

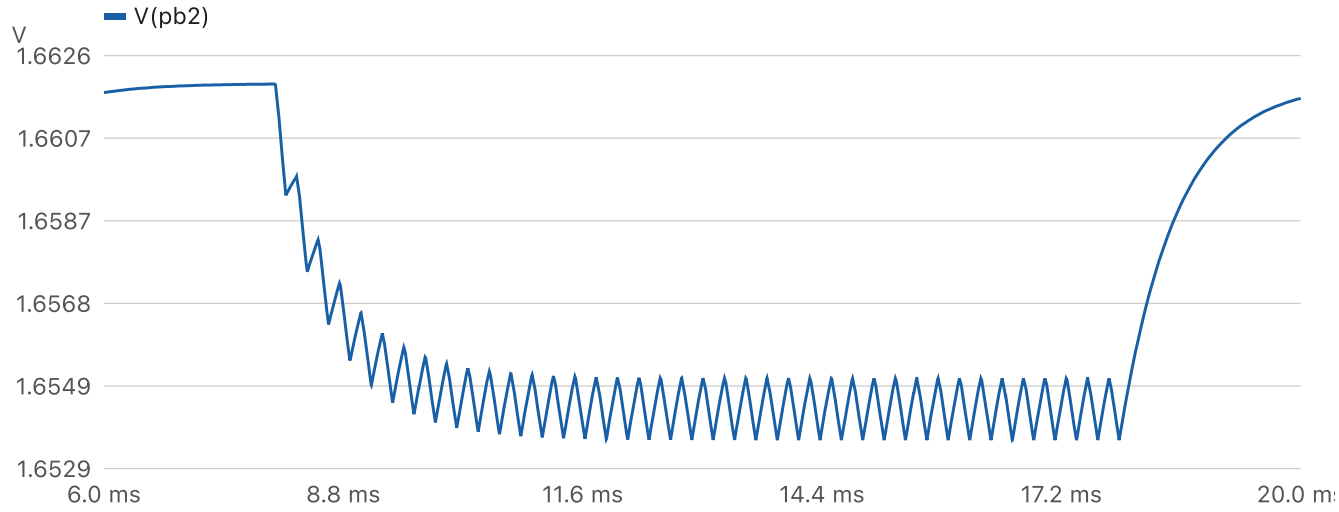
BZ1 · Q3 drain and PB7 (0.3 mH coil)



BZ1 · coil current for 0.1 / 0.3 / 1 mH



BZ1 · VSENSE_5V (PB2) while the buzzer runs, 0.3 mH

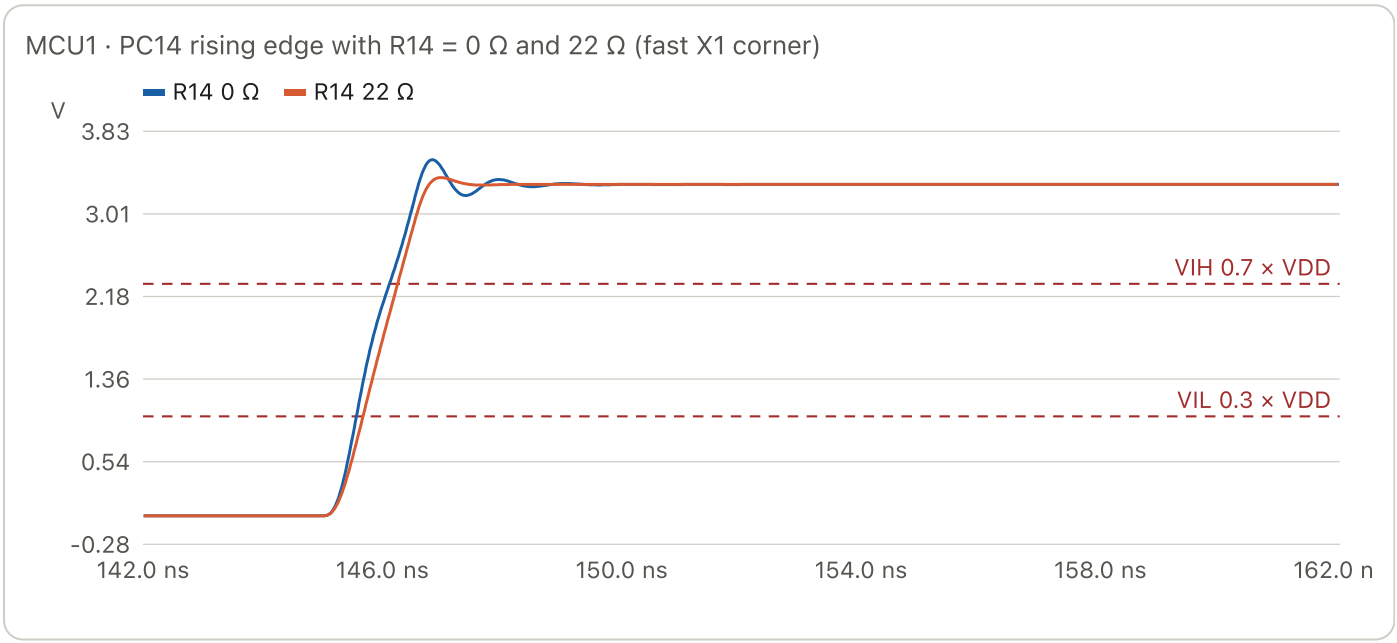


4 · MCU · sheet MCU (page 3)

Parts: X1 SX2M 8 MHz XO → R14 22 Ω → PC14 (HSE bypass), routed lengths from the PCB (HSE_XO 2.29 mm, HSE_IN 2.04 mm, F.Cu over In1 GND, no vias) · NRST: STM32 pull-up R_{PU} 25–55 kΩ, C6 100 nF, 30 mm to the fingers.

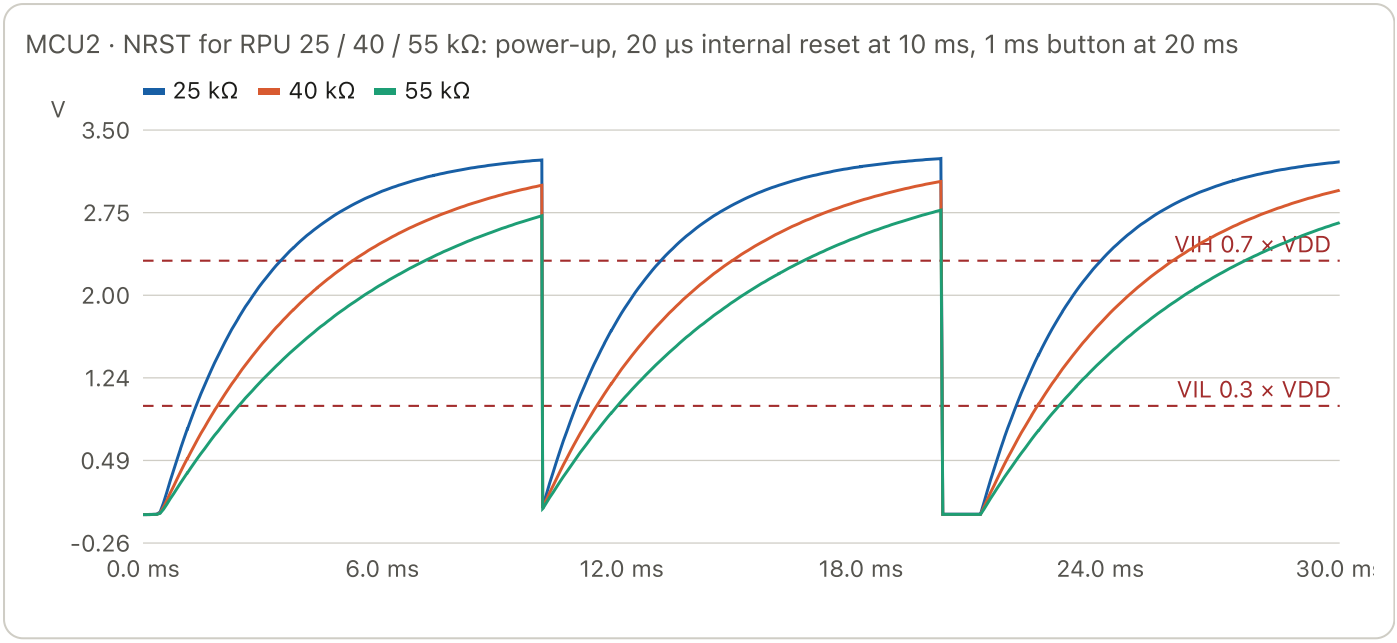
MCU1 · HSE clock edge (`mcu1_hse_clock.net`)

X1's edge rate and output resistance aren't published, so this uses a fast corner: 1.5 ns edges from 15 Ω. The 4.3 mm of trace is far shorter than an edge, so it is lumped with the X1 and LQFP-32 lead inductance (5.2 nH total) into PC14's 5.6 pF. **Without R14**, PC14 rings to 3.55 V and −0.25 V. **With R14 = 22 Ω**, it peaks at 3.37 V and −0.07 V, about 3.5× less ringing, and the 20–80 % edge slows by only 0.03 ns (0.84 → 0.87 ns). Neither case forward-biases PC14's clamp diodes (injection under 1 nA), and the duty cycle at VDD/2 stays 50.0 %, inside the PLL's 45–55 %. R14 does its job.



MCU2 · NRST (`mcu2_nrst.net`)

3V3 rises as in PWR1. The STM32 holds NRST low until VDD passes about 2.0 V (POR / BOR), then R_{PU} charges C6. NRST reaches VIH (0.7 × VDD) **3.0 / 4.9 / 6.7 ms** after POR for R_{PU} 25 / 40 / 55 kΩ. A 20 µs internal reset pulse (driver assumed 50 Ω) pulls C6 down to **53–65 mV**, well below VIL (0.99 V), so ST's 20 µs pulse still reaches the pin through 100 nF. Recovery takes 3.0–6.5 ms. A 1 ms button press takes NRST to ≈ 0 V, with 3.0–6.6 ms recovery.


MCU2 · NRST for RPU 25 / 40 / 55 kΩ: power-up, 20 µs internal reset at 10 ms, 1 ms button at 20 ms
25 kΩ
40 kΩ
55 kΩ
V
3.50
2.75
2.00
1.24
0.49
-0.26
VIH 0.7 × VDD
VIL 0.3 × VDD
0.0 ms
6.0 ms
12.0 ms
18.0 ms
24.0 ms
30.0 m

5 · CAN and M.2 · sheet CAN_M2 (page 4)

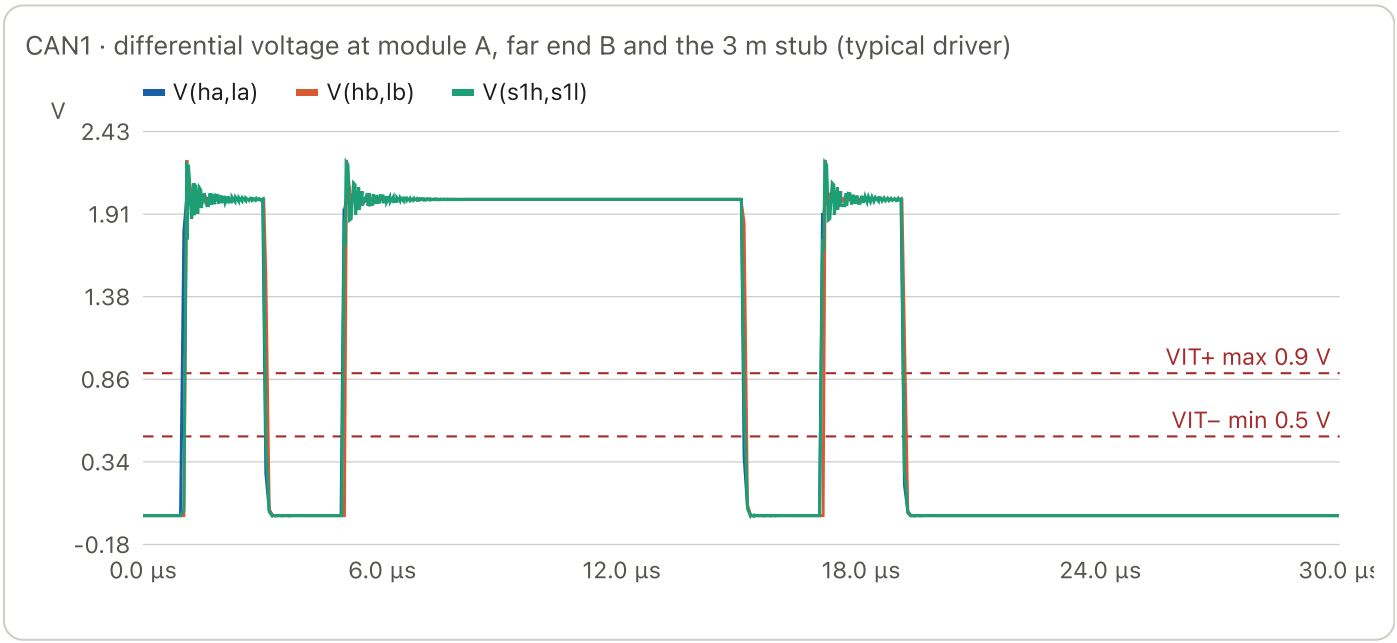
Parts: U3 SN65HVD230 (Rs = 0 Ω, high speed) · D1 NUP2105L · R7 120 Ω + JP1 · fingers 1/3. U3 bus side modelled from TI SLOS346O: dominant VOD 2.0 V typical / 1.5 V minimum into 60 Ω, 35–45 ns differential edges, receiver 35 kΩ to 2.3 V.

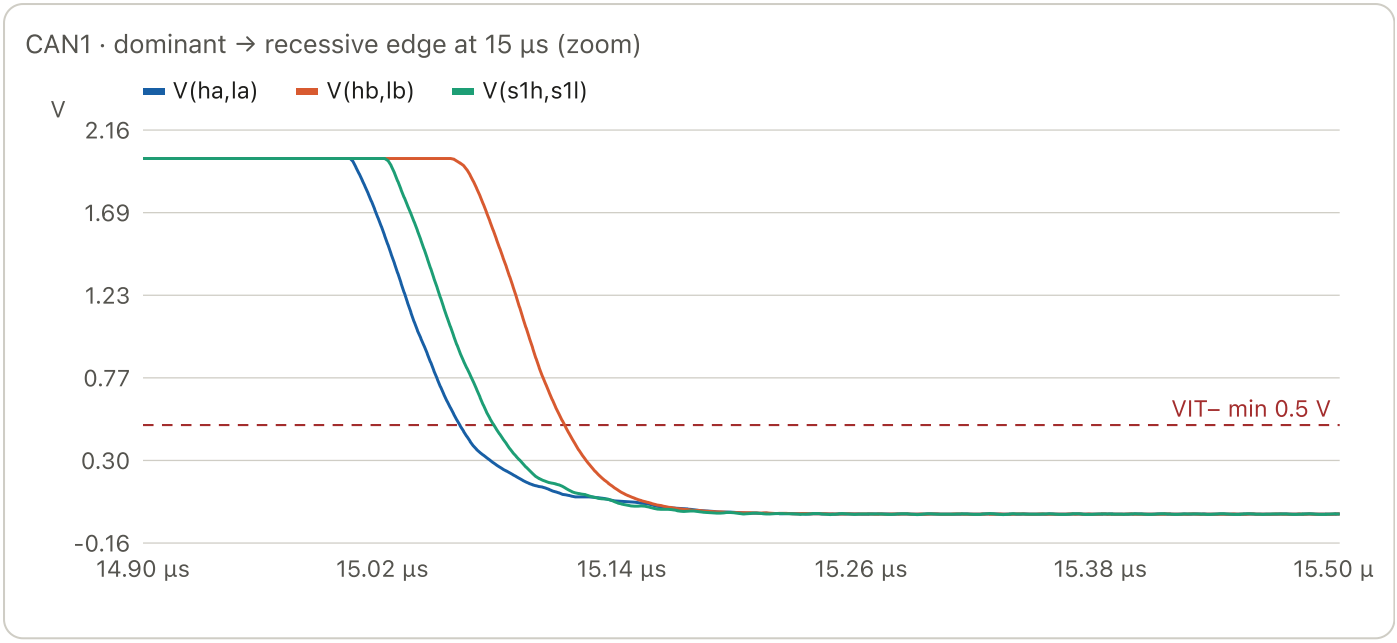
CAN1 · 500 kbit/s on a 10 m bus (`can1_bus_500k.net`)

This module (A, JP1 bridged) transmits D R D D D D D R D R R R R R at 2 µs per bit. Listening are two MES4-MCUs on 0.3 m stubs at 3 m and 6 m (JP1 open) and a terminated MES4-MCU at 10 m (B). Every node has D1 and U3's input capacitance on the bus. The cable is 120 Ω differential at 5 ns/m.

Check	Typical driver	Weakest driver	Limit
Dominant VOD at A / B / stub	2.00 / 2.00 / 2.00 V	1.50 / 1.50 / 1.50 V	≥ 0.9 V (VIT+ max)
Lowest dominant VOD, stub at 3 m	1.90 V	1.47 V	≥ 0.9 V
Ring-back on the recessive edge, stub	2.7 mV	3.7 mV	< 0.5 V (VIT− min)
Recessive VID at the sample point (87.5 %)	≈ 0 V	≈ 0 V	< 0.5 V
TX edge → B crosses 0.9 V / 0.5 V	61 / 106 ns	66 / 88 ns	sample point 1.75 µs

The bus is clean: 0.3 m stubs and the NUP2105L's 26 pF per line are negligible at 500 kbit/s. The edge-to-B times include the 50 ns cable but not U3's own 35–120 ns driver and receiver delays. Even with those added twice for arbitration (t_{LOOP} ≤ 135 ns), the round trip stays under 0.6 µs, far inside the 1.75 µs sample point.





CAN2 · series resistors on CANH / CANL (`can2_series_r.net`)

MES4-CIF rule 8 says "no series resistors on CAN". Here a carrier adds Rs in each line between the fingers and a bus terminated at both ends (60 Ω):

Rs per line	0 Ω	22 Ω	47 Ω	100 Ω
Dominant VOD at the bus, typical driver	2.00 V	1.25 V	0.88 V	0.54 V
Dominant VOD at the bus, weakest driver	1.50 V	1.03 V	0.76 V	0.49 V

22 Ω already falls below the 1.5 V that ISO 11898-2 asks of a transmitter. From 47 Ω, receivers with a 0.9 V threshold can miss dominant bits. Rule 8 is right as written: TVS only, no resistors.

6 · Not simulated, and why

- **I²C1:** the pull-ups are on the carrier. MES4-Carrier-EC MAIN1 covers 400 kHz with 2.2 kΩ and 60 pF (111 ns rise).
- **ESD on CANH / CANL and the fingers:** how the current splits between D1 and U3 depends on U3's internal clamp, which TI doesn't publish. Datasheet ratings: NUP2105L IEC 61000-4-2 ±30 kV contact; SN65HVD230 bus pins ±16 kV HBM.
- **CAN shorted to 24 V:** above U3's 16 V absolute maximum, so destroyed by datasheet (MES4-CIF rule 9). Nothing to simulate.
- **Pin 4 (3V3) shorted:** U2 folds back to 150–350 mA (DRV package), so the module draws ≤ ≈ 0.37 A from +5V. U2 then dissipates up to 1.8 W and cycles in thermal shutdown (180 °C). That's thermal behaviour, not modelled.
- **SWD, VCP, NRST and GPIO fingers:** short, low-speed, carrier-specific lines. USB D+/D– are unused.
- **STM32 internals:** PLL lock, ADC sampling and the exact POR / BOR timing are taken from ST's data, not simulated.

7 · Models and assumptions

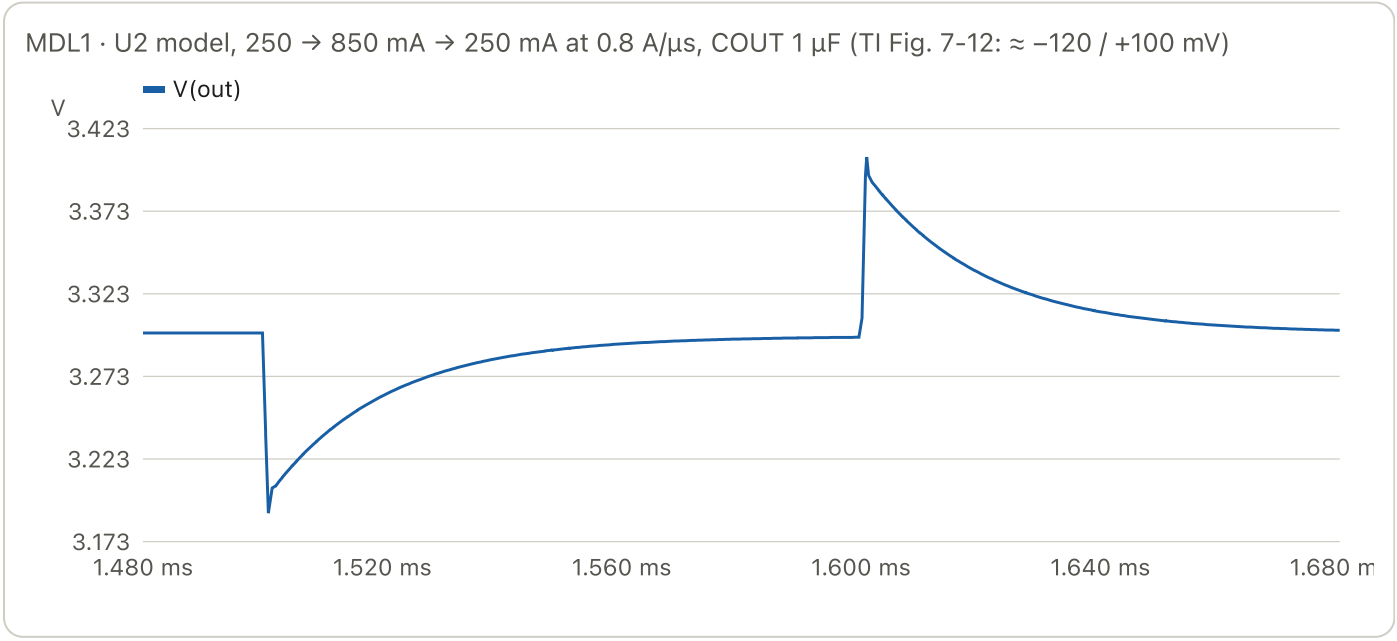
- LTspice 17.2.4 (macOS). Benches and shared models are in `MES4–MCU/sim/`: `models.inc` (device models), `module.inc` (module power network) and `carrier5v.inc` (MES4-Carrier-EC +5V). Run `python3`

`run_sims.py`, then `python3 make_report.py`.

- **U2 TLV76733:** a behavioural model fitted to TI SLVSE84D. UVLO 2.2 V / 130 mV hysteresis, 500 µs soft-start, current limit 1.3 A folding back to 0.25 A, dropout 0.9 Ω, no sinking, PMOS body diode OUT → IN, −60 dB supply ripple above 1 kHz. The control loop (PI, 6 A/V, plus a 0.15 µs pole) is tuned to TI Fig. 7-12. MDL1 below checks the fit: −109 / +105 mV against TI's ≈ −120 / +100 mV. Steps from a near-zero load are worse in silicon (TI Fig. 7-11: about −400 mV from 1 mA), but the module never draws less than about 30 mA from 3V3.
- **U3 SN65HVD230:** bus-side behavioural model. Dominant CANH from VCC − 0.1 V and CANL to 0.75 V through $R_d/2$ each, with R_d = 13.5 Ω (typical, VOD 2.0 V) or 38 Ω (minimum, 1.5 V). VCC current is 10 mA plus the bus current. Driver and receiver propagation delays are not modelled.
- **D2 SMF6.0A:** VBR 7.0 V at 10 mA, R_s 0.16 Ω (10.3 V at 19.4 A), 1.5 nF. **D1 NUP2105L:** two anti-series 29 V junctions per line, 0.5 Ω, 26 pF.
- **Q3 AO3400A and D3 1N4148WS:** the same models as the MES4-Carrier-EC benches. **Q1 / Q2 / Q4 MMBT3904:** 2N3904 parameters. **LEDs:** fitted to the forward voltages in the Power notes (red / yellow 2.0 V at 3 mA, green / white 2.8 V at 2 mA). The LED currents therefore check the transistor stages and the supply range, not the LEDs themselves.
- **Ceramic capacitors** at their DC-bias value: 10 µF 25 V 0805 = 5 µF at 5 V and 6.9 µF at 3.3 V (Power notes). 100 nF parts with 20 mΩ ESR.
- **STM32G0B1:** 20 mA at 64 MHz once VDD > 1.6 V; GPIO output 40 Ω; PC14 5 pF plus a 3 nH lead, clamp diodes to the rails; NRST pull-up 25 / 40 / 55 kΩ, POR / BOR release at about 2.0 V, internal reset driver 50 Ω (assumed).
- **X1 SX2M:** 1.5 ns edges from 15 Ω (fast corner; the datasheet gives neither). **BZ1 KLJ-5018-5040:** 25 Ω coil with inductance stepped 0.1–1 mH; mechanical resonance not modelled.
- **Carrier +5V:** the MES4-Carrier-EC network from its bench PWR5. The URB2405S-6WR3 start-up ramp isn't published; 2 ms and 100 µs are used. Benches with a fixed supply ramp it from 0 V over the first 20 µs so LTspice starts from rest.
- **CAN cable:** two ideal 60 Ω lines at 5 ns/m (120 Ω differential). Common-mode impedance and cable loss are not modelled.

MDL1 · U2 model check (`mdl1_tlv767_fit.net`)

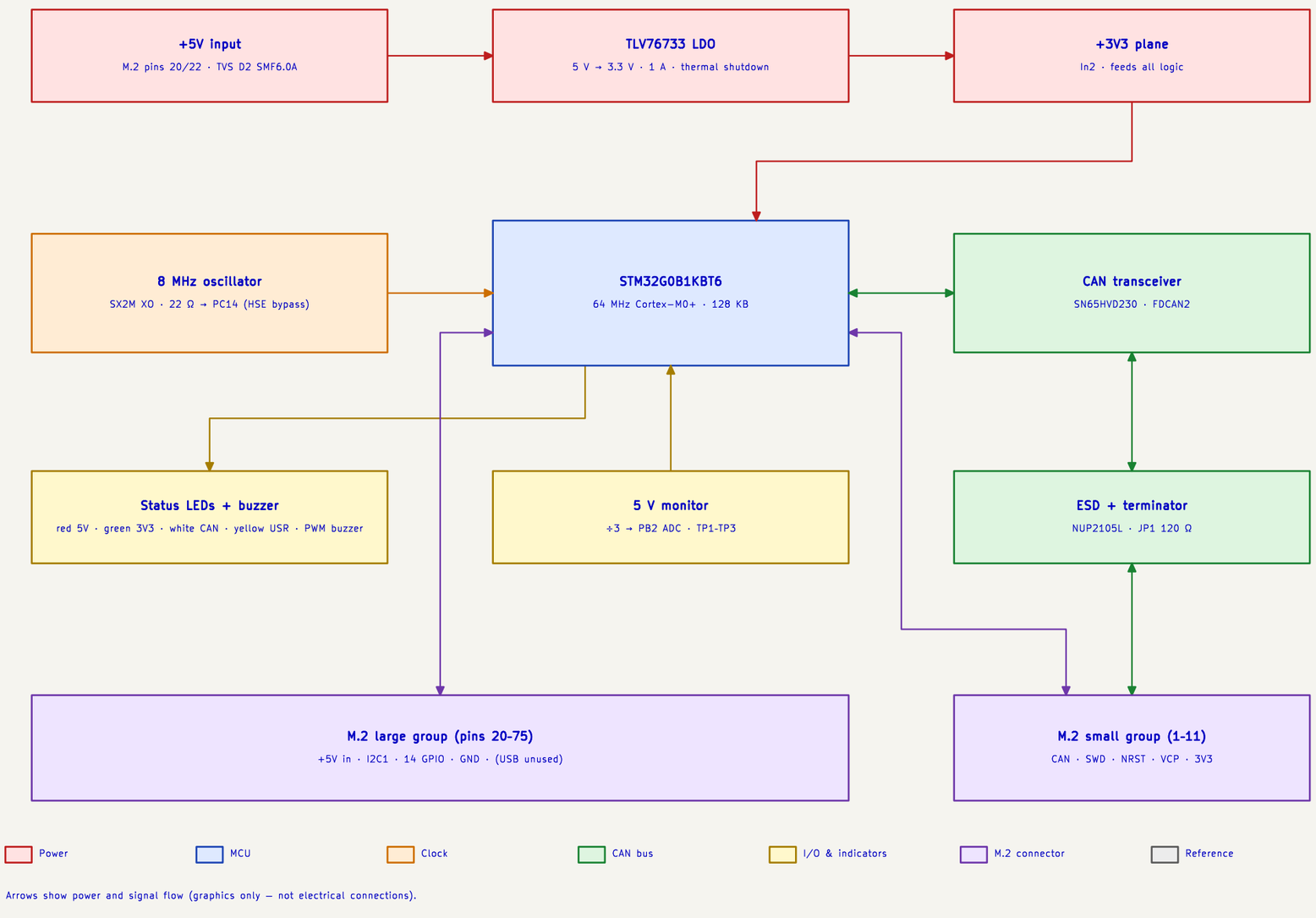
The same test as TI Fig. 7-12: VIN 5 V, COUT 1 µF, 250 → 850 mA in 0.75 µs, then back. The model dips to 3.190 V (−109 mV) and overshoots to 3.405 V (+105 mV), against TI's ≈ −120 / +100 mV. It recovers more slowly than the silicon: about 40 µs, against roughly 10 µs in TI's plot. That makes the PWR5 and BZ1 3V3 figures pessimistic, not optimistic.



MES4-MCU

STM32G0B1 M.2 2242 CAN module · Rev A-d6 · Designer: Aaron · Monkey Robotics

System block diagram



Arrows show power and signal flow (graphics only — not electrical connections).

Hierarchical sheets

Power
File: Power.kicad_sch

Datasheets
File: Datasheets.kicad_sch

MCU
File: MCU.kicad_sch

CAN_M2
File: CAN_M2.kicad_sch

SHEET INDEX

1 Root — overview & block diagram
2 Power — 5 V input, TVS, TLV76733 LDO, power LEDs, status LEDs, buzzer, 5 V monitor, test points
3 MCU — STM32G0B1KBT6, 8 MHz oscillator, decoupling, reset
4 CAN_M2 — SN65HVD230, ESD, termination, M.2 card edge
5 Datasheets — every part with clickable datasheet links

KEY SPECIFICATIONS

Supply ..... +5V in 4.75-5.25 V: ≈ 40 mA typ, ≤ 240 mA worst → carrier ≥ 300 mA/module
Ambient ..... −20...+70 °C (LDO TJ ≤ 92 °C)
3V3 rail ..... TLV76733, 1 A, thermal shutdown; 148 mA worst incl. 50 mA for the carrier
MCU ..... STM32G0B1KBT6, 64 MHz, 128 KB Flash / 144 KB RAM
CAN ..... FDCAN2, 500 kbit/s, CANopen, ESD-protected
Clock ..... 8 MHz XO (±53 ppm worst) → PLL 64 MHz
Protection .... SMF6.0A TVS on +5V, NUP2105L on CAN
LEDs ..... red 5V · green 3V3 · white CAN · yellow USR
Buzzer ..... KLJ-5018 passive 5 V, 90 dB (PB7 4 kHz PWM)
Debug ..... SWD + VCP (USART1) → ST-Link V3MINIE
USB ..... wired to fingers 23/25, not used
Carrier I/O ... 14 GPIO + I2C1 + 3V3 out (≤ 50 mA)
PCB ..... M.2 2242 B-key, 4-layer 0.8 mm JLC04081H-3313, ENIG, bevelled fingers
Layers ..... Top Signal / GND Plane / 3V3 Plane / Bottom Signal
Parts ..... 40 parts: 5 JLC Extended (U1 X1 U2 BZ1 D1), 2 Preferred (U3, D2), rest Basic

REVISION HISTORY (Rev A = not yet fabricated)

d1 2026-10-07 Aaron From-scratch redesign of MES3
d2 2026-10-07 Aaron Datasheet pinout pass, all nets named, Basic LEDs, USB_DN pair
d3 2026-10-07 Aaron Review 1: active buzzer, R14 on HSE, C2/C5 sizes, carrier rules
d4 2026-10-08 Aaron Review 2: TLV76733 LDO, D2 D_SMF, R7 0603, R8 20k, R12 4.7k, R15/R16 at U3, carrier rules, USB unused
d5 2026-10-13 Aaron Smaller buzzer: KLJ-5018-5040 5.5 mm passive, 4 kHz PWM on PB7
d6 2026-10-14 Aaron U1 → STM32G0B1KBT6 (LQFP-32): KBU6 out of stock; review fixes

Carrier spec: MES4-CIF = CARRIER_INTERFACE.md (rev A-d6)

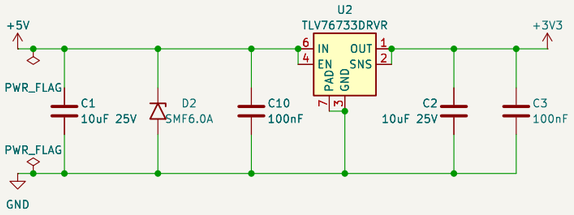
MES4-MCU · STM32G0B1 M.2 2242 CAN module · 4-layer 0.8 mm JLC04081H-3313
Designer: Aaron
Monkey Robotics

Sheet: /
File: MES4-MCU.kicad_sch

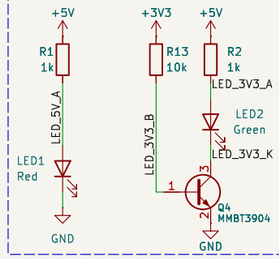
Title: MES4-MCU - Overview

Size: A3	Date: 2026-10-14	Rev: A-d6
KiCad E.D.A. 9.0.2		Id: 1/5

5 V → 3.3 V regulator



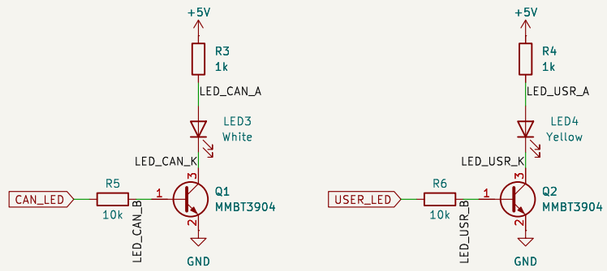
Power indicators



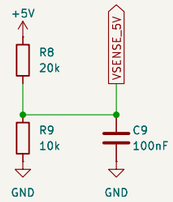
Test points



Status LEDs (5 V, NPN low-side, GPIO driven)



5 V input monitor



Divider (R8 top, R9 bottom):

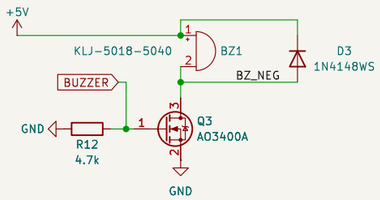
Vsense = V(+5V) × R9 / (R8 + R9)
= 5.0 V × 10k / (20k + 10k)
= 5.0 V × 0.333
= 1.67 V → PB2 (ADC1_IN10)

Full scale = 3.3 V × 3 = 9.9 V
Resolution = 3.3 V / 4096 × 3 ≈ 2.4 mV per count
4.75 V → 1.583 V → ADC ≈ 1965
5.00 V → 1.667 V → ADC ≈ 2068
5.25 V → 1.750 V → ADC ≈ 2172
10.3 V TVS clamp → 3.43 V (< 3.6 V)

C9 filter: source R8 ‖ R9 = 6.7 kΩ
τ = 6.7 kΩ × 100 nF = 0.67 ms
fc = 1 / (2π × τ) ≈ 239 Hz
Divider current = 5 V / 30 kΩ = 0.17 mA

Firmware:
VDDA = 3.0 V × VREFINT_CAL / VREFINT_DATA
V(+5V) = ADC / 4095 × VDDA × 3

Buzzer (KLJ-5018-5040 passive, PWM)



KLJ-5018-5040 = PASSIVE buzzer, no oscillator:
5 V, 25 Ω, ≥ 90 dB @ 4 kHz.

PB7 = TIM4_CH2 PWM 4 kHz, 50 % duty (less = quieter). Idle LOW. Never leave PB7 HIGH: 200 mA DC = 1 W, the coil overheats.

D3: freewheels the coil.
R12 4.7k: Q3 off in reset/boot (weak pull-up on PB7).

DESIGN NOTES & CALCULATIONS — POWER

1. Supply input
   - +5V from the carrier on M.2 pins 20/22 (4.75-5.25 V, max 5.5 V). The carrier supplies +5V only; +3V3 is generated here and is never tied to a carrier rail.
   - D2 SMF6.0A TVS (VRWM 6 V > 5.25 V max; IR ≤ 400 µA at 6 V, far less at 5 V; VC 10.3 V; 200 W; unidirectional: cathode → +5V) clamps spikes on +5V and conducts negative transients to GND. At a 10.3 V clamp: U2 18 V abs, C1 25 V, C10 16 V, Q1/Q2/Q4 40 V, Q3 30 V: safe; PB2 sees 10.3 V × 10k / 30k = 3.43 V via R8/R9: PB2 is FT_ea (VDD + 4 V max), R8 limits < 0.35 mA.
   - C1 10 µF/25 V 0805 (≈ 5 µF at 5 V bias) + C10 100 nF at U2 IN.

2. 3.3 V LDO — U2 TI TLV76733DRVR: VIN 2.5-16 V (18 V abs), 1 A, thermal shutdown 180 °C, fold-back current limit, stable with ≥ 1 µF ceramic, WSON-6 2 × 2 mm (RθJA 77.7 °C/W). EN → IN (always on), SNS → OUT, thermal pad → GND.
   Worst-case 3V3 load:
   STM32G0B1 @ 64 MHz, all peripherals ..... 20 mA
   SN65HVD230 dominant into the 60 Ω bus ..... 67 mA (17 mA no-load + VOD 3 V / 60 Ω; ≈ 17 mA RMS in normal traffic, TI Fig. 1)
   X1 8 MHz oscillator (Icc max) ..... 10 mA
   GPIO + Q4 base drive ..... 1 mA
   Module subtotal ..... 98 mA
   Carrier allowance (M.2 pin 4, ≤ 50 mA) ..... 50 mA
   TOTAL ..... 148 mA (15 % of the 1 A rating)
   Dropout: 1.4 V max @ 1 A (≈ 1.4 Ω) → ≈ 0.2 V @ 148 mA → needs VIN ≥ 3.5 V << 4.75 V OK
   Dissipation: P = (5.25 − 3.3) V × 148 mA = 0.29 W → ΔT = 0.29 W × 77.7 °C/W ≈ 22 °C
   Module rated −20...+70 °C ambient: TJ = 70 + 22 = 92 °C ≤ 125 °C OK.
   A short on pin 4 / TP2 is caught by the fold-back limit + thermal shutdown.
   PCB: thermal pad → top GND copper → vias beside it (none in the pad).
   Output: C2 10 µF/25 V 0805 (≈ 6.9 µF at 3.3 V; TLV767 needs ≥ 0.5 µF effective) + C3 100 nF; C4/C5 at the MCU (MCU sheet).

3. Power indicator LEDs I = (Vsupply − Vf) / R
   LED1 red (+5V present): (5.0 − 2.0) / 1 kΩ = 3.0 mA
   LED2 green (3V3 present): green Vf ≈ 2.8 V is too high to run bright from 3.3 V, so LED2 is fed from +5V and switched by Q4, whose base (R13 10k) is driven by +3V3. It lights once 3V3 is above ≈ 1 V: a 'rail present' indicator, not a 'rail in regulation' check.
   I = (5.0 − 2.8 − 0.1 VCE(sat)) / 1 kΩ ≈ 2.1 mA; base (3.3 − 0.7) / 10 kΩ = 0.26 mA
   All 4 LEDs: 0805, JLC Basic parts, 4 different colours.

4. Status LEDs — +5V supply, MMBT3904 low-side (white Vf ≈ 2.8 V is too high for 3.3 V)
   LED3 white : (5.0 − 2.8 − 0.1 VCE(sat)) / 1 kΩ ≈ 2.1 mA (CAN status)
   LED4 yellow: (5.0 − 2.0 − 0.1) / 1 kΩ ≈ 2.9 mA (user GPIO)
   Base: (3.3 − 0.7) / 10 kΩ = 0.26 mA → IC/IB ≈ 8 (LED3), 11 (LED4) << hFE ≥ 100 → saturated
   CAN_LED = PB4, USER_LED = PB5, GPIO high = LED on.

5. 5 V input monitor — R8 20k / R9 10k / C9 → PB2 (ADC1_IN10)
   VSENSE = 5.0 × 10k / (20k + 10k) = 1.67 V full scale = 3 × VDDA = 9.9 V
   Source impedance 20k ‖ 10k = 6.7 kΩ (< 50 kΩ ADC limit) → sampling time ≥ 39.5 cycles
   Firmware: VDDA = 3.0 V × VREFINT_CAL / VREFINT_DATA
   V(+5V) = ADC / 4095 × VDDA × 3
   Tip: sample with BZ1 silent (its current steps ripple +5V) or average several readings.

6. Buzzer — BZ1 KLJ-5018-5040 PASSIVE magnetic buzzer (no oscillator inside): 5 V (4-6 V), 25 ± 3 Ω coil, ≥ 90 dB @ 10 cm, ≤ 80 mA mean at 4 kHz / 50 % duty; 5.5 × 5.5 × 1.8 mm.
   PB7 = TIM4_CH2 PWM, 4 kHz: PSC 0, ARR 15999, CCR2 8000 (50 %). Lower duty = quieter.
   End each tone with CCR2 = 0 (PB7 LOW). Never DC HIGH: 5 V / 25 Ω = 200 mA = 1 W in the coil.
   Q3 AO3400A low-side (≤ 48 mΩ at VGS 2.5 V); D3 carries the coil current in each PWM off-time.
   R12 4.7k keeps Q3 off in reset/boot even if a weak internal pull-up (≥ 25 kΩ) is on PB7:
   VGS ≤ 3.3 × 4.7 / 29.7 = 0.52 V < 0.65 V (AO3400A VGS(th) min).

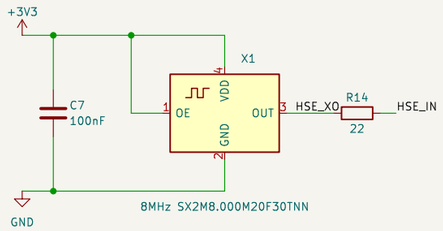
7. Test points (top side, 1 mm pads): TP1 +5V, TP2 +3V3, TP3 GND.

MES4-MCU · STM32G0B1 M.2 2242 CAN module · 4-layer 0.8 mm JLC04081H-3313
Designer: Aaron
Monkey Robotics
Sheet: /Power/
File: Power.kicad_sch

Title: MES4-MCU - Power, LEDs, buzzer & monitoring

Size: A3	Date: 2026-10-14	**Rev: A-d6**
KiCad E.D.A. 9.0.2		Id: 2/5

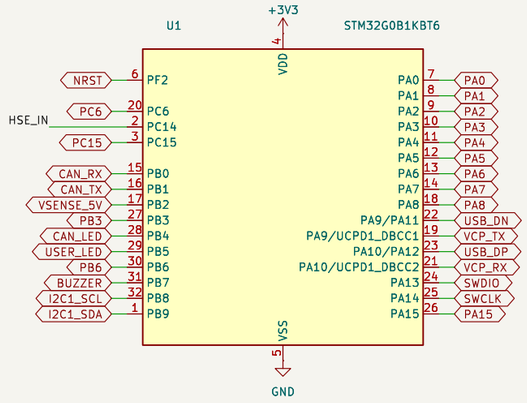
HSE: 8 MHz CMOS oscillator (bypass)



WHY AN OSCILLATOR, NOT A CRYSTAL + 2 CAPACITORS?
- A bare crystal needs the MCU to drive it: OSC_IN + OSC_OUT pins, plus two load caps (CL1/CL2) from each pin to GND.
- On this 32-pin package there is NO OSC_OUT pin (PC15 only has an optional OSC_EN), so a bare crystal cannot be used.
- X1 is an active oscillator: crystal + driver inside the can. It outputs a ready 3.3 V square wave into PC14 (HSE bypass, RCC HSEBYP = 1). Load caps are already inside — none needed.
- C7 100 nF is X1's supply decoupling only (≤ 2 mm from pin 4).
- Pin 1 OE (Tri-State) tied to +3V3 = output always on.

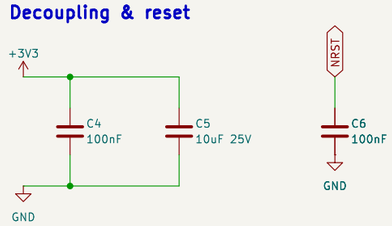
IS 8 MHz ENOUGH? YES — it is only the reference; the PLL multiplies it up to the chip's maximum:
8 MHz ÷ M2 = 4 MHz × N32 = 128 MHz VCO ÷ R2 = 64 MHz SYSCLK (max) ÷ Q2 = 64 MHz FDCAN clock
- CAN 500 kbit/s = 64 MHz ÷ (BRP 8 × 16 tq) — exact. Sample point (1 + 13) ÷ 16 = 87.5 %.
- CAN clock tolerance (seg2 = 2, SJW = 2) ≈ ±0.49 %. X1: ±20 + ±30 (temp) + ±3/yr ≈ ±53 ppm = ±0.005 % → ~90× margin. Internal HSI16 drifts −2.75/+2.0 % → not safe for CAN.
- Layout: X1 → R14 → PC14 ≤ 8 mm, no vias, over solid GND.

STM32G0B1KBT6 (LQFP-32, non-N pinout)



LQFP-32: no exposed pad; pin 5 is the only VSS.
Pin 5 runs to the C4/C5 ground pads and the GND plane.

Decoupling & reset



DECOUPLING (U1 pin 4 = VDD/VDDA, pin 5 = VSS)
- C4 100 nF: supplies the fast current spikes each time the core and I/O switch (MHz). Must sit right at pin 4.
- C5 10 µF 0805 (≈ 6.9 µF at 3.3 V): bulk reservoir for slower load steps (ST AN5096 asks for ≥ 4.7 µF).
- Placement: C4 closest to pin 4 (≤ 2 mm), C5 just behind it; each with its own via straight down to the GND plane.

RESET (NRST, U1 pin 6 → M.2 pin 11)
- NRST is active-low: 0 V = reset, 3.3 V = run. Internal ~40 kΩ pull-up to 3.3 V, so no external resistor is needed.
- C6 100 nF to GND is a noise filter: the NRST trace runs to the M.2 fingers / ST-Link cable and picks up spikes and ESD. A spike couples in through only a few pF, so against 100 nF it barely moves NRST: no accidental reset. After a real reset the 40 kΩ pull-up recharges C6 in ≈ 5 ms (3-7 ms).
- Real resets (ST-Link, carrier button NRST → GND) hold the pin low long enough to drain C6. C6 also debounces the button.
- Everything on NRST may only pull LOW (open-drain, wired-OR). Never drive it high, never add more capacitance — the STM32's own ~20 µs reset pulse must still pull it fully low.
- C6 ≤ 2 mm from pin 6, short via to GND. (MES4-CIF §4a)

DESIGN NOTES & CALCULATIONS — MCU

1. U1 STM32G0B1KBT6 — LQFP-32 (0.8 mm), non-N pinout (PA9/PA10 on 19/21, PA11/PA12 on 22/23).
   Order exactly STM32G0B1KBT6, NOT the ...KBT6N variant (pins 17, 20, 26-29 differ; VDDIO2 on pin 20).
   Footprint: KiCad LQFP-32_7x7mm_P0.8mm (IPC), checked against the DS13560 LQFP32 drawing.
   No exposed pad: pin 5 is the only VSS pin.

2. Decoupling (ST AN5096): C4 100 nF + C5 10 µF 0805 (≈ 6.9 µF at 3.3 V) at VDD pin 4.
   C6 100 nF on NRST (internal pull-up ≈ 40 kΩ) filters reset glitches.

3. Clock tree
   X1 8 MHz CMOS XO (±20 + ±30 ppm, duty 45-55 %, OE high) → R14 22 Ω → PC14 (HSE bypass).
   32-pin package: PC14 = only HSE input, no OSC_OUT (PC15 OSC_EN = optional AF; GPIO here) → no crystal.
   PLL: input = 8 MHz / M(2) = 4 MHz (50 % duty; PLL needs 45-55 %) VCO = 4 MHz × N(32) = 128 MHz
   SYSCLK = VCO / R(2) = 64 MHz FDCAN kernel = VCO / Q(2) = 64 MHz
   HSE line X1 → R14 → PC14: ≤ 8 mm, no vias, 0.3 mm spacing; R14 22 Ω at X1 OUT (less ringing/EMI).

4. CAN bit timing — FDCAN2, 500 kbit/s classic (CANopen)
   f_bit = 64 MHz / (BRP 8 × (1 + TSEG1 13 + TSEG2 2)) = 64 MHz / 128 = 500 kbit/s
   Sample point = (1 + 13) / 16 = 87.5 % (CANopen recommendation). SJW = 2
   Oscillator ≈ ±53 ppm worst << CAN limit 0.485 % OK (HSI16 drifts −2.75/+2.0 %: not usable for CAN)

5. Pin allocation
   PB0 / PB1 ..... FDCAN2 RX / TX → U3 PB4 / PB5 ... CAN_LED / USER_LED
   PA9 / PA10 .... USART1 TX / RX → ST-Link VCP PB7 ..... BUZZER (TIM4_CH2 PWM)
   PA11 / PA12 ... USB DN / DP (wired, unused) PB2 ..... VSENSE_5V (ADC1_IN10, ÷3)
   PA13 / PA14 ... SWDIO / SWCLK(BOOT0) PB8 / PB9 .. I2C1 SCL / SDA (carrier monitors)
   PF2 ..... NRST PC14 ..... HSE_IN
   To carrier (14 GPIO): PA0-PA8, PA15, PB3, PB6, PC6, PC15 (ADC-capable: PA0-PA7)

6. Boot, programming & debug
   PA14 = SWCLK and BOOT0. Option bytes: nBOOT_SEL = 1, nBOOT0 = 1 (check/set once in STM32CubeProgrammer)
   → always boots from flash; the PA14 pin is NOT read as BOOT0 → SWCLK stays free and a connected debugger can never change the boot mode.
   No BOOT0 button (it would fight the ST-Link's SWCLK driver). Flash/debug = ST-Link over SWD.
   A blank chip starts the ST bootloader; after the first SWD flash, power-cycle once (the empty-flash flag is only re-read at power-on).
   ST bootloader when needed: firmware jumps to system memory on a VCP command, or clear nBOOT0 via SWD → use the bootloader → set nBOOT0 = 1 again.
   ST system bootloader (AN2606): USART1 PA9/PA10 (8E1, via ST-Link VCP). It also probes USART2 PA2/PA3, SPI1 PA4-PA7, USB PA11/PA12 and I2C1 PB6/PB7 (pulled up: PB6 = no gate drive).
   No FDCAN bootloader on this package (needs PD0/PD1).
   VCP default: USART1 115200 8N1 → ST-Link V3MINIE (STDC14 pins 13/14).

7. Monitoring (firmware)
   VDDA = 3.0 V × VREFINT_CAL / VREFINT_DATA (factory calibration at 3.0 V)
   V(+5V) = ADC(PB2) / 4095 × VDDA × 3 (R8 20k / R9 10k)
   Unused GPIO → analog mode (lowest power, best EMC).

U1 PIN MAP — what every STM32G0B1KBT6 pin does

Pin	Port	Net / signal	Function	Connected to	M.2 pin	ADC	Spare alternate functions	Notes
1	PB9	I2C1_SDA	I2C	M.2 finger 53 (carrier INA226)	53	—	FDCAN1_TX, TIM4_CH4, USART3_RX	Pull-up on carrier to M.2 pin 4 (+3V3)
2	PC14	HSE_IN	CLOCK	X1 OUT via R14 22 Ω	—	—	—	8 MHz HSE bypass; X1→R14→PC14 ≤ 8 mm
3	PC15	PC15	GPIO	M.2 finger 51 (spare input)	51	—	TIM15_BK	Weak pin: sink ≤ 3 mA, ≤ 2 MHz, never source
4	VDD/VDDA	+3V3	POWER	U2 LDO out; C4 100 nF + C5 10 µF	4	—	—	Also the ADC reference → full scale = 3.3 V
5	VSS	GND	POWER	GND plane	GND	—	—	—
6	PF2-NRST	NRST	RESET	C6 100 nF; ST-Link STDC14 pin 12	11	—	—	Active low, internal pull-up; pull LOW only
7	PA0	PA0	GPIO	M.2 finger 49 (relay 1 FB)	49	IN0	TIM2_CH1, USART4_TX, SPI2_SCK	Wake-up pin WKUP1
8	PA1	PA1	GPIO	M.2 finger 47 (relay 2 FB)	47	IN1	TIM2_CH2, USART4_RX, SPI1_SCK	—
9	PA2	PA2	GPIO	M.2 finger 45 (relay 3 FB)	45	IN2	TIM2_CH3, TIM15_CH1, USART2_TX	—
10	PA3	PA3	GPIO	M.2 finger 43 (relay 4 FB)	43	IN3	TIM2_CH4, TIM15_CH2, USART2_RX	—
11	PA4	PA4	GPIO	M.2 finger 41 (ADS1232 DOUT)	41	IN4	DAC1_OUT1, SPI1_NSS, TIM14_CH1	DAC capable; TT pin: 3.6 V max
12	PA5	PA5	GPIO	M.2 finger 39 (ADS1232 SCLK)	39	IN5	DAC1_OUT2, SPI1_SCK, TIM2_CH1	DAC capable; TT pin: 3.6 V max
13	PA6	PA6	GPIO	M.2 finger 37 (ADS1232 PDWN)	37	IN6	SPI1_MISO, TIM3_CH1, TIM16_CH1	—
14	PA7	PA7	GPIO	M.2 finger 35 (relay 4 drive)	35	IN7	SPI1_MOSI, TIM3_CH2, TIM17_CH1	—
15	PB0	CAN_RX	CAN	U3 SN65HVD230 R (pin 4)	—	—	—	FDCAN2_RX
16	PB1	CAN_TX	CAN	U3 D (pin 1) + R15 10k pull-up	—	—	—	FDCAN2_TX
17	PB2	VSENSE_5V	ADC	R8 20k / R9 10k ÷3 + C9 from +5V	—	IN10	—	5 V → 1.67 V at pin; V(5V) = 3 × reading
18	PA8	PA8	GPIO	M.2 finger 31 (relay 1 drive)	31	—	TIM1_CH1, RCC_MCO, SPI2_NSS	May read LOW in reset until UCPD strobe (FW)
19	PA9	VCP_TX	UART	M.2 6 → ST-Link T_VCP_TX	6	—	USART1_TX	ST-Link VCP + bootloader UART
20	PC6	PC6	GPIO	M.2 finger 29 (relay 2 drive)	29	—	TIM3_CH1, TIM2_CH3, LPUART2_TX	—
21	PA10	VCP_RX	UART	M.2 8 ← ST-Link T_VCP_RX	8	—	USART1_RX	ST-Link virtual COM; bootloader UART
22	PA11 [PA9]	USB_DN	USB	M.2 finger 25	25	—	USB_DM	USB D− - not used; keep PA11_RMP = 0
23	PA12 [PA10]	USB_DP	USB	M.2 finger 23	23	—	USB_DP	USB D+ - not used; keep PA12_RMP = 0
24	PA13	SWDIO	DEBUG	M.2 7 → ST-Link STDC14 pin 4	7	—	—	Do not reuse as GPIO
25	PA14-BOOT0	SWCLK	DEBUG	M.2 9 → ST-Link STDC14 pin 6	9	—	—	Shared with BOOT0 — see MCU notes
26	PA15	PA15	GPIO	M.2 finger 63 (INA226 ALERT)	63	—	TIM2_CH1, SPI1_NSS, USART2_RX	—
27	PB3	PB3	GPIO	M.2 finger 61 (relay 3 drive)	61	—	TIM1_CH2, TIM2_CH2, SPI1_SCK, I2C2_SCL	—
28	PB4	CAN_LED	LED	R5 → Q1 → LED3 (white)	—	—	TIM3_CH1	HIGH = LED on (PWM dimming possible)
29	PB5	USER_LED	LED	R6 → Q2 → LED4 (yellow)	—	—	TIM3_CH2	HIGH = LED on (PWM dimming possible)
30	PB6	PB6	GPIO	M.2 finger 59 (spare, BL pull-up)	59	—	TIM1_CH3, TIM4_CH1, USART1_TX, I2C1_SCL	—
31	PB7	BUZZER	OUT	Q3 gate (R12 4.7k pull-down) → BZ1	—	—	TIM4_CH2	Passive buzzer: 4 kHz PWM; idle LOW, never DC
32	PB8	I2C1_SCL	I2C	M.2 finger 55 (carrier INA226)	55	—	FDCAN1_RX, TIM4_CH3, TIM16_CH1	Pull-up on carrier to M.2 pin 4 (+3V3)

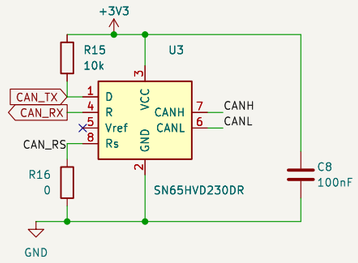
Totals: 14 carrier GPIOs (PA0-PA7 also ADC IN0-IN7) + I2C1 + VCP UART + SWD + NRST on the M.2 (USB wired, unused); on-module: CAN, 2 LEDs, buzzer, 5 V sense, HSE.
Internal ADC channels (no pin needed): VREFINT → measures the real 3V3 rail, temperature sensor. [PA9] / [PA10] = SYSCFG remap option, leave off.

PIN RESTRICTIONS (ST DS13560)
(!) PA8 may read LOW during reset/boot: UCPD dead-battery pull-down, set by PA9's level. Released when firmware strobes (F1).
(!) PC15 RTC-domain pin: sinks ≤ 3 mA, ≤ 2 MHz, ≤ 30 pF; must NOT source current (no LED).
(!) PA4/PA5 are IT pins: 3.6 V max (not 5 V tolerant). PA6/PB3: no negative injection.
(!) PA9/PA10 = UCPD DBCC pins: a HIGH level here enables PA8's dead-battery pull-down (F1).
(!) U1 pins 22/23 are named 'PA9/PA11', 'PA10/PA12' in ST's symbol: remappable pads, used as PA11/PA12. Leave the remap off.

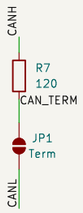
FIRMWARE RULES
F1 SystemInit(): set SYSCFG_CFGR1.UCPD1_STROBE and UCPD2_STROBE first (frees PA8).
F2 USB unused: keep the USB peripheral and its D+ pull-up (USB_BCDR.DPPU_DPD) OFF.
F3 Production: verify nBOOT_SEL = 1, nBOOT0 = 1; power-cycle after the first flash. RDP1 blocks the ST bootloader (updates = mass-erase + reflash, or a custom IAP). Never RDP2.
F4 Buzzer: TIM4 PWM only; idle LOW, never DC.
F5 VSENSE: buzzer silent, ≤ 10 kSps, × 3.
F6 Clock: HSEBYP while HSEON = 0, then HSEON; PLLM = 2, PLLN = 32, PLLR = 2, PLLQ = 2.
F7 CSS on, in the CSS NMI set PB1 (CAN_TX) to input first (R15 = recessive), then re-init.
F8 FDCAN NBTP = value − 1: NBRP 7, NTSEG1 12, NTSEG2 1, NSJW 1.
F9 Never enable LSE / PC15 AF0-1; keep NRST bidirectional; enable BOR.

MES4-MCU · STM32G0B1 M.2 2242 CAN module · 4-layer 0.8 mm JLC04081H-3313
Designer: Aaron
Monkey Robotics
Sheet: /MCU/
File: MCU.kicad_sch
Title: MES4-MCU - MCU
Size: A3 | Date: 2026-10-14 | **Rev: A-d6**
KiCad E.D.A. 9.0.2 | Id: 3/5

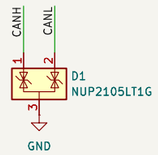
CAN transceiver (3.3 V)



Bus termination (bridge JP1 at bus ends)



Bus ESD protection



WHY 120 Ω AT EACH END OF THE BUS? (R7 + JP1)
- A CAN cable is a twisted pair with ≈ 120 Ω characteristic impedance. A bit that reaches an open (unterminated) end bounces back as a reflection and rings on top of the next bits → bit / CRC errors.
- A 120 Ω resistor at BOTH physical ends absorbs the wave: no reflection. Two in parallel → 120 ‖ 120 = 60 Ω measured between CANH and CANL.
- Recessive (= 1): no node drives; the 60 Ω termination holds CANH and CANL at the same voltage (Vdiff ≈ 0 V). The ≈ 2.3 V common-mode level comes from the transceivers' own bias, not from the resistors. Dominant (= 0): U3 drives CANH up / CANL down, Vdiff ≈ 2 V into 60 Ω.

WHY A SOLDER JUMPER (JP1)?
- Every module carries R7, but only the 2 nodes at the cable ends may terminate. JP1 is OPEN by default → R7 is out of circuit. Bridge JP1 (solder blob) only on the two END modules.
- Wrong count: 1 terminator = 120 Ω → reflections, errors at 500 kbit/s
  3 terminators = 40 Ω → drivers overloaded, small signal
- R7 power: 2^2 / 120 ≈ 33 mW typ, 3^2 / 120 = 75 mW max in use (0603 = 100 mW OK)

CHECK WITH A MULTIMETER (all power OFF, measure CANH ↔ CANL):
- ≈ 60 Ω → correct, both ends terminated
- ≈ 120 Ω → only one end terminated (bridge the other JP1)
- ≤ 40 Ω → too many terminators (open the extra JP1s)
- open → no termination at all

LAYOUT: CANH/CANL routed as a pair (0.10 / 0.25 mm, ≈ 118 Ω diff), R7/JP1 and D1 close to the fingers; keep branch cables (stubs) short.

DESIGN NOTES & CALCULATIONS — CAN & M.2

1. U3 SN65HVD230 — 3.3 V CAN transceiver, 500 kbit/s (CANopen); C8 100 nF at VCC.
   Rs (pin 8) → R16 0 Ω → GND = high-speed mode. Fit ≈ 10 kΩ instead of R16 for slower, lower-EMI edges (+≈ 50 ns, negligible at 500 kbit/s). Vref (pin 5) unused.
   D (pin 1): internal 100 kΩ pull-up + R15 10 kΩ to +3V3 → CAN_TX idles recessive while the MCU is in reset, blank or being flashed.
   Bus pins rated −4…+16 V, common mode −2…+7 V; no TXD time-out (carrier rule 9).

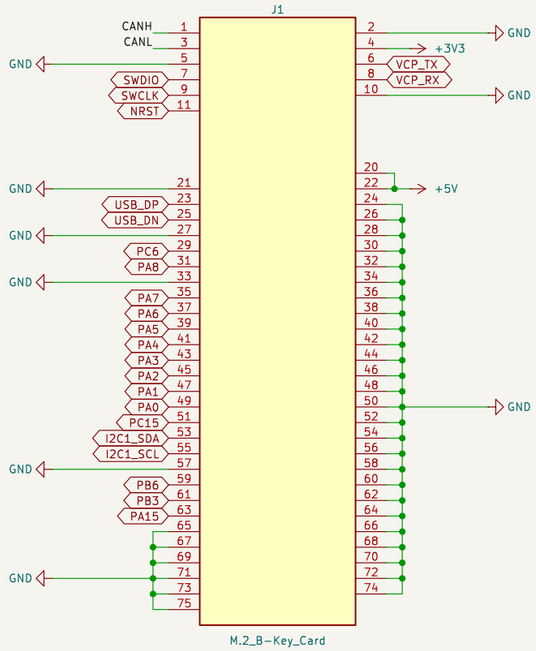
2. Bus termination — 120 Ω at each physical bus end → 60 Ω total
   JP1 OPEN by default; bridge only on a module at a bus end.
   R7 0603 (100 mW): 33 mW typ (VOD 2 V), 75 mW worst (VOD 3 V, bus held dominant) OK
   Check: power off, CANH-CANL ≈ 60 Ω on a correctly terminated bus.

3. D1 NUP2105L — dual-line CAN ESD protector between U3 and the fingers (stand-off 24 V, breakdown ≥ 26.2 V, 26-30 pF). It absorbs ESD / fast surges only. It does NOT protect U3 against a DC fault between +16 V and +26 V, or below −4 V → carrier rule 9.

4. PCB impedance — JLC04081H-3313 (F.Cu over In1 GND, prepreg 0.0994 mm, εr 4.1)
   CAN 0.10 / 0.25 mm pair → ≈ 118 Ω diff (bus 120 Ω)
   USB 0.15 / 0.13 mm pair → ≈ 90 Ω diff (USB wired to fingers 23/25, not used)
   (Hammerstad-Jensen; verify with JLCPCB's impedance calculator before ordering.)

5. M.2 2242 B-key card edge: custom pinout, odd fingers top / even fingers bottom; see the finger map tables. Card: 0.8 mm, ENIG, bevelled gold fingers.
   Carrier spec: MES4-CIF (CARRIER_INTERFACE.md).

M.2 2242 B-key card edge



CARRIER MUST-FOLLOW RULES (the module depends on them; full text in MES4-CIF §0)
1 CUSTOM pinout: NOT a standard M.2 card. Use only on MES carriers.
2 +5V only on pins 20/22: 4.75-5.5 V, ≥ 300 mA per module (worst case ≈ 240 mA). Fingers 2/70/72/74 are GND here: never put 3.3 V on them.
3 Pin 4 = the module's 3V3 OUTPUT: never drive it; carrier loads ≤ 50 mA.
4 Input protection is the carrier's job: reverse polarity, fuse or e-fuse, cable surge, +5V never below −0.3 V. The module only has the D2 TVS.
5 No hot-plug: power off before inserting or removing a module.
6 No back-powering: never drive a module pin (VCP, GPIO, I²C, NRST) while pin 4 is unpowered.
7 I²C1 pull-ups to pin 4 (+3V3) must always be fitted.
8 ESD at every external connector. CAN: low-capacitance TVS ONLY, NO series resistors (they weaken the dominant level). GPIO/SWD/NRST: TVS + optional 22-100 Ω. USB fingers 23/25: not used, leave unconnected.
9 CAN: CANH/CANL never share a connector or cable with a supply above 12 V; all nodes on one solid ground (common mode −2…+7 V); exactly two 120 Ω terminators at the cable ends; stubs ≤ 0.3 m at 500 kbit/s.
10 All I/O ≤ 3.3 V (PA4/PA5 3.6 V abs). PA8 may read LOW at reset · PC15 must not source current · NRST may only be pulled LOW.

J1 M.2 FINGER MAP — what every edge-connector pin carries

TOP SIDE (F.Cu) — odd fingers

Pin	Signal	Type	Goes to (on module)
SMALL FINGER GROUP 1-11 · CAN + debug			
1	CANH	CAN	U3 CANH, D1 ESD, R7
3	CANL	CAN	U3 CANL, D1 ESD, JP1
5	GND	GND	GND plane
7	SWDIO	DEBUG	U1 PA13 (pin 24)
9	SWCLK	DEBUG	U1 PA14-BOOT0 (pin 25)
11	NRST	RESET	U1 NRST (pin 6) + C6
▲▼ KEY NOTCH · 13-19 = no fingers (B-key slot) ▲▼			
LARGE FINGER GROUP 21-75 · USB, GPIO, I2C, GND			
21	GND	GND	GND plane
23	USB_DP	USB	U1 PA12 (pin 23) - unused
25	USB_DN	USB	U1 PA11 (pin 22) - unused
27	GND	GND	GND plane
29	PC6	GPIO	U1 PC6 (pin 20)
31	PA8	GPIO	U1 PA8 (pin 18) - low@reset
33	GND	GND	GND plane
35	PA7	GPIO	U1 PA7 (pin 14) · ADC IN7
37	PA6	GPIO	U1 PA6 (pin 13) · ADC IN6
39	PA5	GPIO	U1 PA5 · IN5 · 3.6 V max
41	PA4	GPIO	U1 PA4 · IN4 · 3.6 V max
43	PA3	GPIO	U1 PA3 (pin 10) · ADC IN3
45	PA2	GPIO	U1 PA2 (pin 9) · ADC IN2
47	PA1	GPIO	U1 PA1 (pin 8) · ADC IN1
49	PA0	GPIO	U1 PA0 (pin 7) · ADC IN0
51	PC15	GPIO	U1 PC15 (pin 3) - weak
53	I2C1_SDA	I2C	U1 PB9 (pin 1)
55	I2C1_SCL	I2C	U1 PB8 (pin 32)
57	GND	GND	GND plane
59	PB6	GPIO	U1 PB6 (pin 30)
61	PB3	GPIO	U1 PB3 (pin 27)
63	PA15	GPIO	U1 PA15 (pin 26)
65	GND	GND	GND plane
67	GND	GND	GND plane
69	GND	GND	GND plane
71	GND	GND	GND plane
73	GND	GND	GND plane
75	GND	GND	GND plane

BOTTOM SIDE (B.Cu) — even fingers

Pin	Signal	Type	Goes to (on module)
SMALL FINGER GROUP 2-10 · +3V3, VCP UART, GND			
2	GND	GND	GND plane
4	+3V3	POWER	U2 LDO out (≤ 50 mA out)
6	VCP_TX	UART	U1 PA9 TX → ST-Link
8	VCP_RX	UART	U1 PA10 RX ← ST-Link
10	GND	GND	GND plane
▲▼ KEY NOTCH · 12-18 = no fingers (B-key slot) ▲▼			
LARGE FINGER GROUP 20-74 · +5V in, GND			
20	+5V	POWER	5 V IN → U2 LDO, D2 TVS
22	+5V	POWER	5 V IN → U2 LDO, D2 TVS
24	GND	GND	GND plane
26	GND	GND	GND plane
28	GND	GND	GND plane
30	GND	GND	GND plane
32	GND	GND	GND plane
34	GND	GND	GND plane
36	GND	GND	GND plane
38	GND	GND	GND plane
40	GND	GND	GND plane
42	GND	GND	GND plane
44	GND	GND	GND plane
46	GND	GND	GND plane
48	GND	GND	GND plane
50	GND	GND	GND plane
52	GND	GND	GND plane
54	GND	GND	GND plane
56	GND	GND	GND plane
58	GND	GND	GND plane
60	GND	GND	GND plane
62	GND	GND	GND plane
64	GND	GND	GND plane
66	GND	GND	GND plane
68	GND	GND	GND plane
70	GND	GND	GND plane
72	GND	GND	GND plane
74	GND	GND	GND plane

Blue = small finger group (1-11) · Orange = key notch (12-19, no fingers) · Green = large finger group (20-75)
39 GND fingers: every even finger 24-74 is GND, under the odd signal fingers. Carrier: +5V on 20 + 22 only, ≥ 300 mA per module.
Pin 4 is the module's +3V3 OUTPUT (ST-Link VTref, ≤ 50 mA) — never drive it from the carrier. USB 23/25: wired, not used.
All 14 GPIOs are 3.3 V logic (PA0-PA7 also analog in; PA4/PA5 3.6 V max). Carrier rules: box on this sheet + MES4-CIF.

MES4-MCU · STM32G0B1 M.2 2242 CAN module · 4-layer 0.8 mm JLC04081H-3313
Designer: Aaron
Monkey Robotics
Sheet: /CAN_M2/
File: CAN_M2.kicad_sch

Title: MES4-MCU - CAN & M.2

Size: A3	Date: 2026-10-14	**Rev: A-d6**
KiCad E.D.A. 9.0.2		Id: 4/5

DATASHEETS & REFERENCE DOCUMENTS

Every part in the BOM with its manufacturer datasheet and LCSC page (JLCPCB assembly part numbers).

Ref	Qty	Part number	Manufacturer	Function	LCSC	Datasheet (click)
U1	1	STM32G0B1KBT6	STMicroelectronics	MCU · Cortex-M0+ 64 MHz · 128 KB flash · LQFP-32 (NOT the ...T6N variant)	C5270165	DS13560 (ST)
↳ U1 docs		RM0444	STMicroelectronics	Reference manual: RCC/HSE bypass, FDCAN, option bytes (nBOOT_SEL)	-	RM0444
↳ U1 docs		AN2606	STMicroelectronics	System bootloader: USART1 PA9/PA10 (no FDCAN on this package)	-	AN2606
↳ U1 docs		AN5096	STMicroelectronics	STM32G0 hardware getting started: decoupling, NRST, clocks	-	AN5096
↳ U1 docs		Product page	STMicroelectronics	STM32G0B1KB: all documents, errata, CAD models	-	Product page
U2	1	TLV76733DRVR	Texas Instruments	LDO 3.3 V 1 A · VIN 2.5-16 V · thermal shutdown · WSON-6 2×2	C2848334	TLV767 (TI)
U3	1	SN65HVD230DR	Texas Instruments	3.3 V CAN transceiver · SOIC-8	C12084	SN65HVD230 (TI)
X1	1	SX2M8.000M20F30TNN	SCTF	8 MHz CMOS oscillator · ±20 ppm (+±30 ppm temp) · 2.5 × 2.0 mm	C2901594	SX2M series (LCSC)
D1	1	NUP2105LT1G	onsemi	Dual-line CAN bus ESD protector · SOT-23	C14486	NUP2105L (onsemi)
D2	1	SMF6.0A	Hongjiacheng	TVS 6 V 200 W unidirectional · SMF / SOD-123FL	C19077499	SMF series (LCSC)
D3	1	1N4148WS	Changjing (CJ)	Switching diode 100 V 150 mA · SOD-323	C2128	1N4148WS (LCSC)
Q1, Q2, Q4	3	MMBT3904	Changjing (CJ)	NPN transistor 40 V 200 mA · SOT-23	C20526	MMBT3904 (LCSC)
Q3	1	AO3400A	Alpha & Omega	N-MOSFET 30 V 5.7 A · SOT-23	C20917	AO3400A (AOS)
BZ1	1	KLJ-5018-5040	Changzhou Keliking (KELIKING)	Passive buzzer 5 V, 25 Ω, 4 kHz PWM, ≥ 90 dB · 5.5 × 5.5 × 1.8 mm	C18186309	KLJ-5018-5040 (LCSC)
LED1	1	NCD0805R1	NationStar	LED red · 0805	C84256	NCD0805R1 (LCSC)
LED2	1	KT-0805G	Hubei KENTO	LED emerald green · 0805	C2297	KT-0805G (LCSC)
LED3	1	KT-0805W	Hubei KENTO	LED white · 0805	C34499	KT-0805W (LCSC)
LED4	1	KT-0805Y	Hubei KENTO	LED yellow · 0805	C2296	KT-0805Y (LCSC)
C1, C2, C5	3	CL21A106KAYNNNE	Samsung	MLCC 10 µF 25 V X5R · 0805	C15850	CL21A106KAYNNNE (LCSC)
C3, C4, C6, C7, C8, C9, C10	7	CL05B104KO5NNNC	Samsung	MLCC 100 nF 16 V X7R · 0402	C1525	CL05B104KO5NNNC (LCSC)
R1, R2, R3, R4	4	0402WGF1001TCE	UNI-ROYAL	Resistor 1 kΩ 1 % 62.5 mW · 0402	C11702	0402WGF1001TCE (LCSC)
R5, R6, R9, R13, R15	5	0402WGF1002TCE	UNI-ROYAL	Resistor 10 kΩ 1 % 62.5 mW · 0402	C25744	0402WGF1002TCE (LCSC)
R7	1	0603WAF1200T5E	UNI-ROYAL	Resistor 120 Ω 1 % 100 mW · 0603 (CAN termination)	C22787	0603WAF1200T5E (LCSC)
R8	1	0402WGF2002TCE	UNI-ROYAL	Resistor 20 kΩ 1 % 62.5 mW · 0402	C25765	0402WGF2002TCE (LCSC)
R12	1	0402WGF4701TCE	UNI-ROYAL	Resistor 4.7 kΩ 1 % 62.5 mW · 0402	C25900	0402WGF4701TCE (LCSC)
R14	1	0402WGF220JTCE	UNI-ROYAL	Resistor 22 Ω 1 % 62.5 mW · 0402 (HSE series)	C25092	0402WGF220JTCE (LCSC)
R16	1	0402WGF0000TCE	UNI-ROYAL	Resistor 0 Ω · 0402 (U3 Rs link)	C17168	0402WGF0000TCE (LCSC)
J1	-	M.2 2242 B-key card edge	(PCB gold fingers)	Not assembled: PCB feature. Carrier uses an M.2 B-key socket, e.g. LCSC C590859	-	socket (LCSC)
JP1, TP1-TP3	-	PCB features	-	Not assembled: solder jumper (CAN termination) and test pads	-	-
MES4-CIF	-	CARRIER_INTERFACE.md	Monkey Robotics	Carrier interface spec, rev A-d6 (rules also on sheet 4); file in the project folder	-	-

How to use: blue entries are hyperlinks. In KiCad hover/click the link; in the exported PDF just click it.
Each symbol's own Datasheet field holds the same link, so selecting a part and pressing D opens its datasheet.
LCSC links open the LCSC part page (stock, price, footprint); LCSC datasheet links open LCSC's PDF viewer.
Pinouts of U1, U2, U3, X1, D1, D2, Q1-Q4, BZ1 were checked against these datasheets (last full re-check 2026-10-13). Every symbol also carries MPN + Manufacturer fields.

MES4-MCU · STM32G0B1 M.2 2242 CAN module · 4-layer 0.8 mm JLC04081H-3313
Designer: Aaron
Monkey Robotics
Sheet: /Datasheets/
File: Datasheets.kicad_sch
Title: MES4-MCU - Datasheets & references
Size: A3 | Date: 2026-10-14 | **Rev: A-d6**
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